



# Boards & Beyond: Pulmonary Slides

Color slides for USMLE Step 1 preparation  
from the Boards and Beyond Website

Jason Ryan, MD, MPH

2020 Edition

Boards & Beyond provides a virtual medical school curriculum used by students around the globe to supplement their education and prepare for board exams such as USMLE Step 1.

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# Pulmonary Embryology

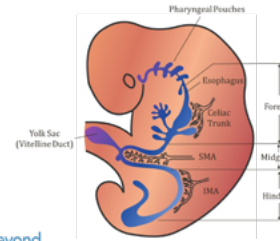
## Pulmonary Embryology

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## Lung Embryology

- Lung bud ("respiratory diverticulum")
  - Outgrowth of **foregut** (future esophagus)
  - Forms during **4<sup>th</sup> week of development**



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## Lung Maturation

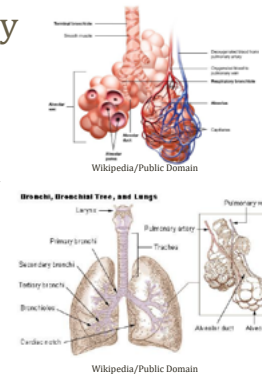
Stages/Periods

- Psuedoglandular (5-16wk)
- Canalicular (16-26wk)
- Saccular (26wk-birth)
- Alveolar (after birth)

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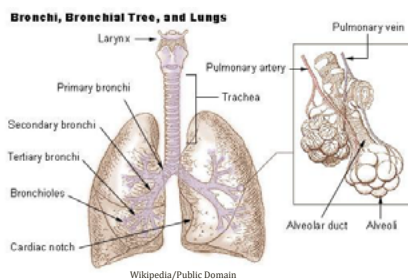
## Lung Anatomy

- Bronchi
  - Hyaline cartilage
- Bronchioles
  - No cartilage
  - Terminal → respiratory
- Alveoli
  - Capillaries
  - Gas exchange



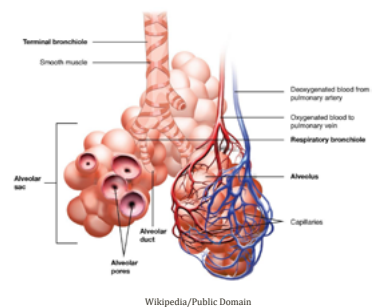
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## Lung Anatomy



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## Lung Anatomy

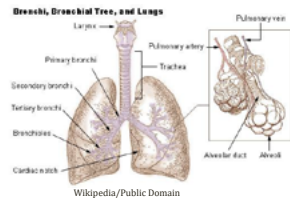


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## Pseudoglandular Period

5-16 weeks

- Lungs resemble a gland
- Branching to level of terminal bronchioles
- No respiratory bronchioles or alveoli present



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## Fetal Respiration

- **Fetal breathing** movements occur in utero
- Baby **aspirates amniotic fluid**
- Stimulates **lung development**
- Growth of respiratory muscles
- Important for growth during pseudoglandular phase

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## Fetal Respiration

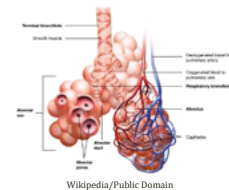
- **Oligohydramnios:**
  - Pulmonary hypoplasia
  - Part of Potter's sequence
  - Caused by fetal kidney abnormalities

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## Canalicular Period

16-26 weeks

- Terminal bronchioles divide
- Form respiratory bronchioles
- Respiratory bronchioles divide into alveolar ducts
- Survival after birth possible at end of period

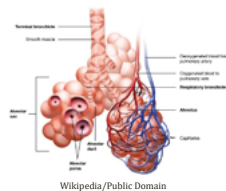


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## Canalicular Period

16-26 weeks

- Airway lumens become larger
- **Type II pneumocytes** form
  - Produce surfactant
  - Lowers surface tension
  - Keeps alveoli open

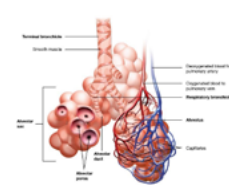


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## Saccular Period

26 weeks - birth

- Terminal sacs (primitive alveoli) form
- Capillaries multiply in contact with alveoli



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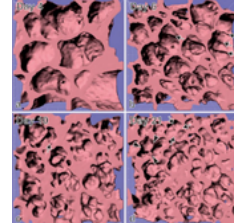
## Alveolar Period

After birth

- At birth, only about 1/3 of alveoli present
- Following birth:
  - ↑ number of respiratory bronchioles and alveoli
- **Continued lung development** through age 10

## Alveolarization

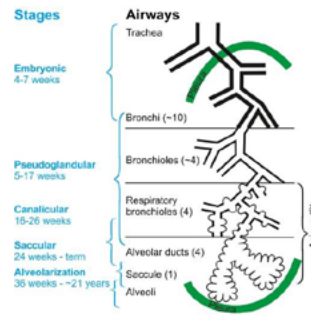
- Airspaces subdivided
- New walls formed (septa)



Johannes Schittny, Cell Tissue Res. 2017; 367 (3) 427

## Bronchopulmonary Dysplasia

- Occurs in premature infants
- Treated in NICU
- Surfactant, oxygen, mechanical ventilation
- Oxygen toxicity and lung trauma
- **Alveolarization** does not progress normally
- Respiratory problems during infancy
- Often improves during childhood



Johannes Schittny, Cell Tissue Res. 2017; 367 (3) 427

## Pulmonary Hypoplasia

- Oligohydramnios (Potter's sequence)
- Congenital diaphragmatic hernia
  - Defective formation **pleuroperitoneal membrane**
  - Hole in diaphragm
  - Abdominal organs herniate into chest
  - In utero herniation → **pulmonary hypoplasia**
  - Often fatal

## Bronchogenic Cysts

- Abnormal budding of **foregut**
- Usually found in mediastinum
- Contain clear fluid
  - Air seen when infected

## Bronchogenic Cysts

- Do not communicate with lungs
- Lined by **respiratory epithelium**
  - Columnar, ciliated
- Walls contain **cartilage** (diagnostic criteria)
- Often asymptomatic
- May lead to pneumonia, compression of airway

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The Radiology Assistant

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## Pulmonary Vascular Resistance

- In utero
  - **PVR is high**
  - Canalicular stage: few/no pulmonary capillaries
  - Later stages: hypoxemia → vasoconstriction
  - Umbilical venous blood:  $\text{PaO}_2$  30mmHg;  $\text{O}_2\text{sat}$ =80%
  - Only about 10% of cardiac output to lungs
- **At birth**
  - **PVR falls** significantly
  - 100% cardiac output through lungs

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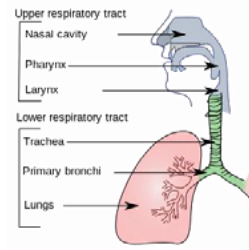
# Pulmonary Anatomy

## Pulmonary Anatomy

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## Respiratory Tract

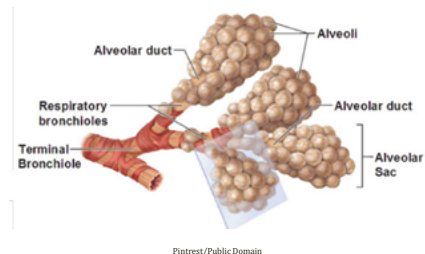


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## Zones

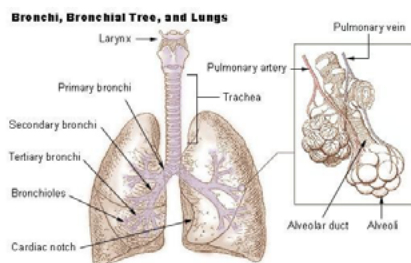
- **Conducting Zone**
  - No gas exchange
  - Large airways, nose, pharynx, trachea, bronchi
  - Filters, warms, humidifies air
  - Anatomic dead space
- **Respiratory Zone**
  - Gas exchange
  - Respiratory bronchioles, alveolar ducts, alveoli

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## Lung Anatomy



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## Bronchi and Bronchioles

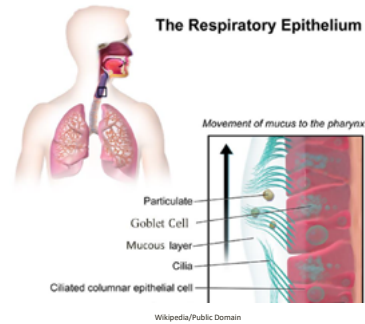
- **Bronchi (cartilage)**
  - Primary (left and right)
  - Secondary/lobar
  - Tertiary/segmental
- **Bronchioles (no cartilage)**
  - Lobular/large
  - Terminal
  - Respiratory (feeds alveoli)

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## Airway Cells

- **Goblet cells**
  - Secrete **mucus**
  - Mostly glycoproteins and water
  - Protects against particulates, infection
- **Ciliated epithelial cells**
  - Beating cilia move mucus to epiglottis
  - Mucus swallowed
- **Club cells (bronchioles)**
  - Non-ciliated epithelial cells
  - Secrete protective proteins
  - Detoxification (P450 enzymes)

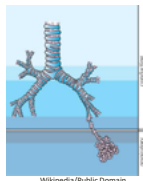
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## Respiratory Epithelium

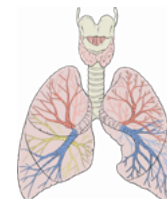
- **Trachea and bronchi**
  - Ciliated **pseudostratified columnar** epithelial cells
  - Goblet cells
- **Bronchioles**
  - Epithelium transitions
  - Forms ciliated **simple cuboidal** epithelium
  - Club cells (terminal bronchioles)



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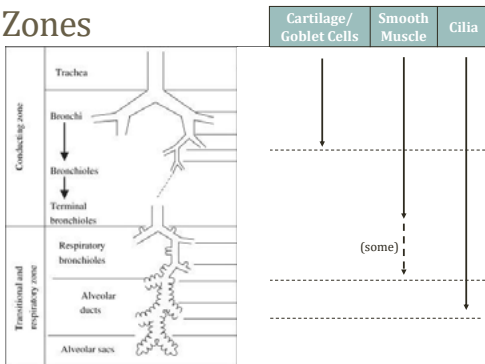
## Smooth Muscle

- **Conducting airway** walls contain smooth muscle
- Sympathetic activation (beta-2)
  - Bronchodilation
- Parasympathetic activation (M3)
  - Bronchoconstriction



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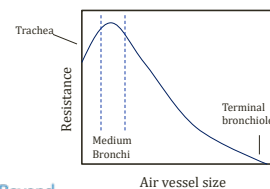
## Zones



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## Resistance to Air Flow

- Upper airways about 50% resistance
  - Nose, mouth, pharynx
- Lower airway resistance
  - Highest in **medium bronchi** (turbulent flow)
  - Lowest in terminal bronchioles - slow laminar flow



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## Alveoli

- Small sacs
- Separated by septa
- Simple squamous epithelium (pneumocytes)
- Gas exchange
- Surrounded by capillaries



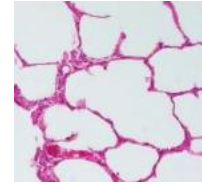
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## Pneumocytes

Alveolar Epithelial Cells

- Type 1
  - Most common (97% of cells)
  - Thin for gas exchange
- Type 2
  - Produce **surfactant**
  - Can proliferate to form other cell types
  - Key for **regeneration** after injury
- Alveolar macrophages

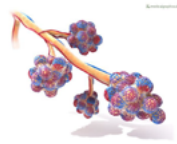


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## Surfactant

- Exhale → alveoli shrink
- Collapse → atelectasis
- ↓ efficiency gas exchange
- Surfactant prevents collapse of alveoli



Medical Graphics/Public Domain

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## Surfactant

- Secreted by **type 2 pneumocytes**
- Mix of **lecithins** (lipid substance)
- Especially **dipalmitoylphosphatidylcholine**



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## Surface Tension

- Alveoli lined with film of liquid
- Liquid-liquid forces shrink surface area into sphere
- **Surface tension** = liquid-liquid forces

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## Law of Laplace

- Determines collapsing pressure
  - Forces tending to collapse alveoli
  - Low collapsing pressure = easy to remain open
  - High collapsing pressure = difficult to remain open

$$\text{Collapsing Pressure} = \frac{2 * (\text{surface tension})}{\text{radius}}$$

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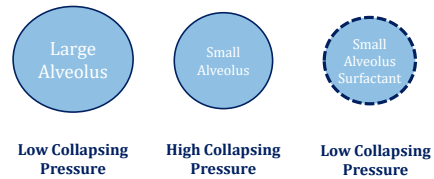
## Law of Laplace

- Lungs contain many small alveoli
- Small radius = high distending pressure
- Need low surface tension to remain open
- Surfactant **reduces surface tension**
- Increases lung compliance (less stiff, more floppy)

$$\text{Collapsing Pressure} = \frac{2 * (\text{surface tension})}{\text{radius}}$$

## Law of Laplace

$$\text{Collapsing Pressure} = \frac{2 * (\text{surface tension})}{\text{radius}}$$



## Fetal Lung Maturity

- Lungs **“mature”** when adequate surfactant present
- Occurs around **35 weeks**
- Lecithin-sphingomyelin ratio (L/S ratio)
- Both produced equally until ~35 weeks
- Ratio >2.0 in amniotic fluid suggests lungs mature



## Neonatal RDS

Neonatal Respiratory Distress Syndrome

- **Surfactant deficiency**
- High surface tension
- Atelectasis
- Decreased lung compliance
- Hypoxemia/↑ pCO<sub>2</sub> (poor ventilation)
- Poorly responsive to O<sub>2</sub>
  - Lungs collapsed (alveoli)
  - Intrapulmonary shunting

## Neonatal RDS

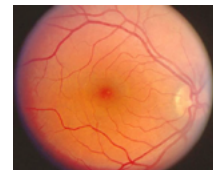
Risk Factors

- Prematurity
- Maternal diabetes
  - High insulin levels decrease surfactant production
- Cesarean delivery
  - Baby spared stress response at delivery
  - Reduced fetal cortisol
  - Reduction in surfactant

## Neonatal RDS

Complications

- Bronchopulmonary dysplasia
  - Oxygen toxicity
  - **Alveolarization** does not progress normally
  - Respiratory problems during infancy
- Patent ductus arteriosus
  - Hypoxia keeps shunt open
- Retinopathy of prematurity
  - Oxygen → free radical formation
  - Neovascularization in the retina
  - Retinal detachment → blindness





## Neonatal RDS

### Prevention and Treatment

- Preterm delivery: **betamethasone**
  - Corticosteroid
  - Given to mother to stimulate surfactant production
- Treatment: surfactant
  - Administered via endotracheal tube



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## Lobes

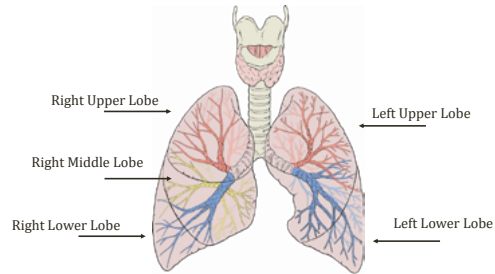
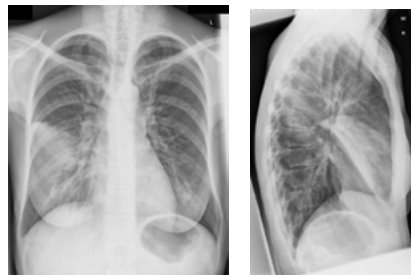


Image courtesy of Patrick J. Lynch, medical illustrator

## Right Upper Lobe



## Right Middle Lobe



## Right Lower Lobe



## Left Upper Lobe



## Left Lower Lobe



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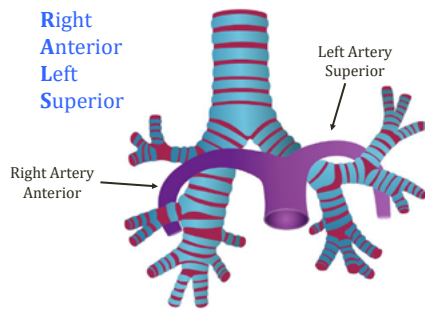
## Foreign Body Aspiration

- Commonly occurs in children (peanuts)
- **Right lung** is more common site of aspiration
  - Right bronchus wider with less angle
  - More vertical path to lung
- **Right lung:** 60% cases
  - Majority in main bronchus
  - Small number in right lower lobe bronchus
- **Left lung:** 23% cases
  - Majority in main bronchus
  - Small number in left lower lobe bronchus

Source: Eren et al. Foreign body aspiration in children: experience of 1160 cases. Ann Trop Paediatr. 2003;23(1):31.

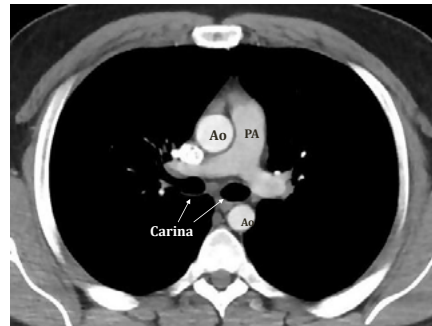
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## Mediastinal Anatomy



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## Mediastinal Anatomy



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## Mediastinal Structures

- Mediastinum: space between lungs
- Divided into 3 anatomical compartments
  - Anterior
  - Middle
  - Posterior
- Differential diagnosis of mass varies by compartment

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## Mediastinal Compartments



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## Mediastinal Structures

| Compartment | Major Structures                                | Masses                                                        |
|-------------|-------------------------------------------------|---------------------------------------------------------------|
| Anterior    | Thymus, internal mammary arteries, lymph nodes  | Thyroid, Thymic neoplasm, Teratoma, Lymphoma                  |
| Middle      | Pericardium, heart, aorta, airway and esophagus | Lymphadenopathy: lymphoma, sarcoid, or metastatic lung cancer |
| Posterior   | Spine, nerves and spinal ganglia                | Neurogenic tumors: schwannoma, neuroblastoma                  |

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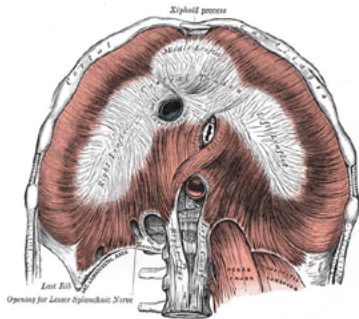
## Anterior Mediastinal Masses

Terrible Ts

- **Thymic masses**
  - Half of anterior masses derive from thymus
  - Thymoma: associated with myasthenia gravis
- Teratoma or **germ cell tumors** in adults
  - Mediastinum: most common location extra nodal GCT
  - Teratomas, seminomas
- Terrible **lymphomas**
- **Thyroid** growths
  - Enlarged or ectopic thyroid tissue may present as mass
  - Usually connected to thyroid gland

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## Diaphragm



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## Diaphragm

- Caval opening
  - T8
  - Inferior vena cava
- Esophageal hiatus
  - T10
  - Esophagus, Vagus nerve
- Aortic hiatus
  - T12
  - Aorta, thoracic duct, azygous vein

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## Diaphragm

- Innervated by **C3, C4, C5 (phrenic nerve)**
- Diaphragm irritation → “referred” shoulder pain
  - Classic example: gallbladder disease
  - Also lower lung masses
  - Irritation can cause dyspnea and hiccups
- Cut nerve → diaphragm elevation, dyspnea
  - “Paradoxical movement” → Moves up with inspiration
  - Can see on fluoroscopy (“sniff test”)

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## Muscles of Quiet Respiration

- Diaphragm → inspiration
- Exhalation is passive with normal (“quiet”) breathing



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## Exercise Breathing

- Inspiration (neck)
  - Scalenes – raise ribs
  - Sternocleidomastoids – raise sternum
- Exhalation (abdomen)
  - Rectus muscle
  - Internal/external obliques
  - Transverse abdominis
  - Internal intercostals
- Use of accessory muscles in respiratory distress

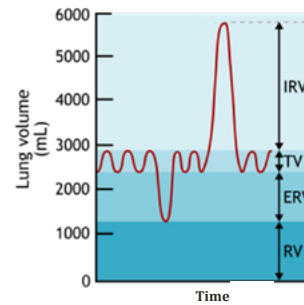
# Pulmonary Physiology

## Pulmonary Physiology

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## Lung Volumes



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Image courtesy of Michal Komorniczak, Medical Illustrations

## Lung Volumes

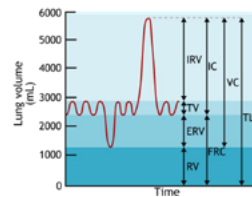
- Tidal volume (TV)
  - In/out air with each quiet breath
- Expiratory reserve volume (ERV)
  - Extra air pushed out with force beyond TV
  - RV remains in lungs
- Inspiratory reserve volume (IRV)
  - Extra air can be drawn in with force beyond TV
  - Lungs filled to capacity
- Residual volume (RV)
  - Air that can't be blown out no matter how hard you try

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## Lung Capacities

Capacity = sum of two volumes

- Total lung capacity
  - Sum of all volumes
  - $RV + ERV + IRV + TV$
- Inspiratory capacity
  - Most air you can inspire
  - $TV + IRV$
- Vital capacity
  - Most you can exhale
  - $TV + IRV + ERV$

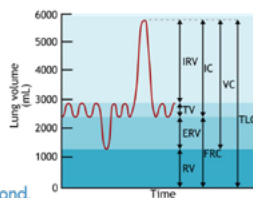


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## Lung Capacities

Capacity = sum of two volumes

- **Functional Residual Capacity**
  - Residual volume after quiet expiration
  - $RV + ERV$
  - Volume when system is relaxed
  - Equilibrium: chest wall pulling out = lungs pulling in



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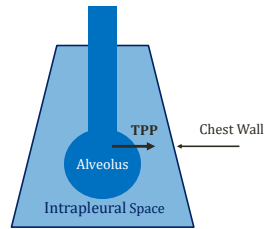
## Lung Pressures

- Atmospheric pressure = 760 mmHg = 0 mmHg
- Alveolar pressure = pressure within alveoli
- Intrapleural pressure = pressure in pleural space
- **Transpulmonary pressure**
  - Alveolar pressure - intrapleural pressure
  - Pressure across walls of alveoli
  - Necessary to keep alveoli open

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## Transpulmonary Pressure

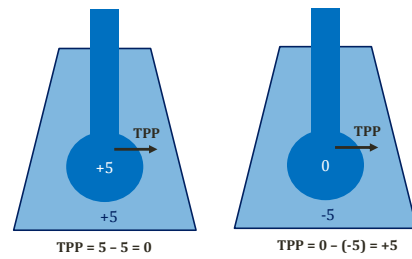
- Alveolar Pressure – Intrapleural Pressure



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## Transpulmonary Pressure

- Alveolar Pressure – Intrapleural Pressure



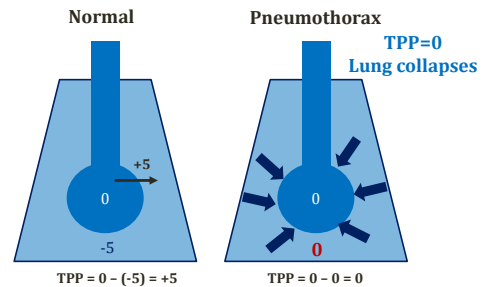
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## Intrapleural Pressure

- **Negative** during normal quiet breathing
- Alveoli and lungs tend to **collapse**
  - Pull inward/recoil
  - Need outward force to keep walls open
- Chest wall tends to **expand**
  - Spring outward
  - Creates negative pressure in pleural space
- Negative pressure “sucks” alveoli open

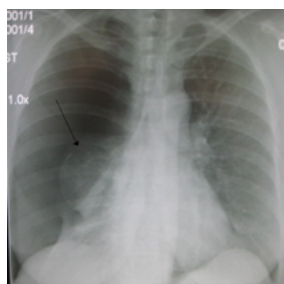
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## Pneumothorax



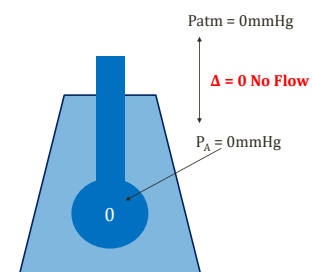
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## Pneumothorax

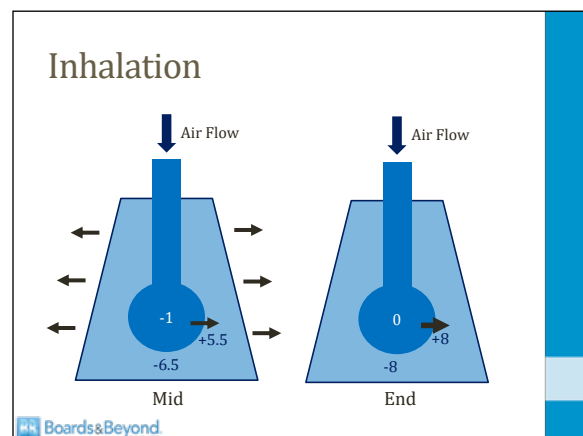
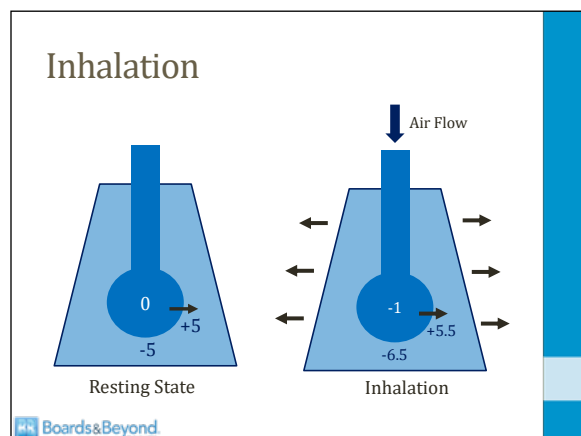
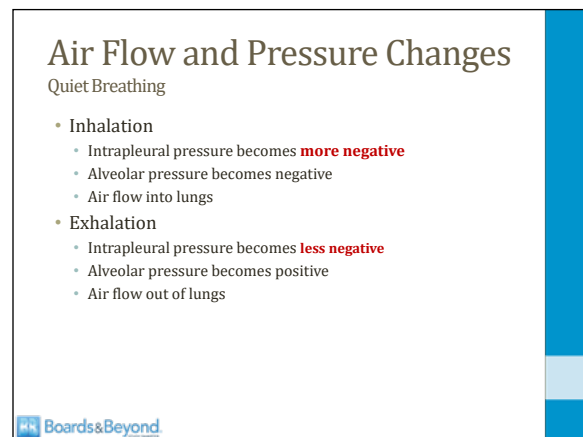
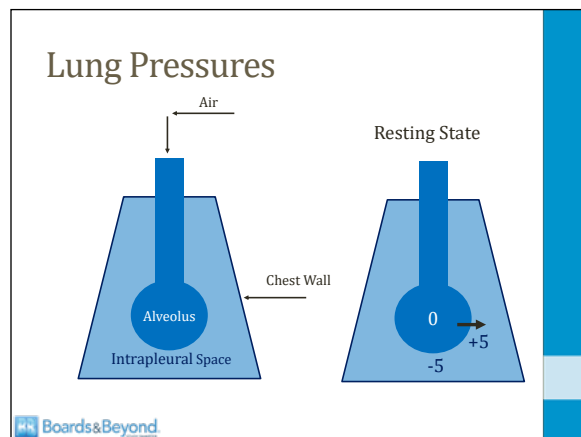
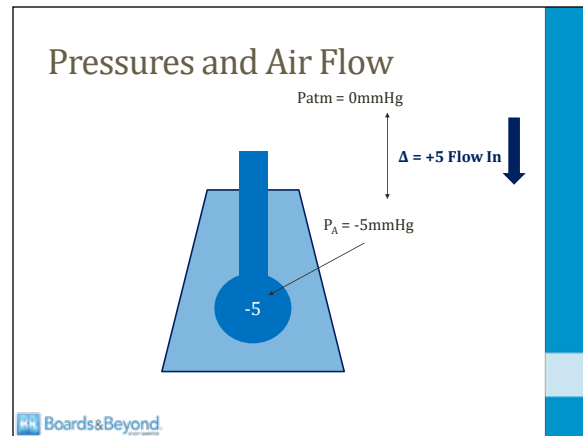
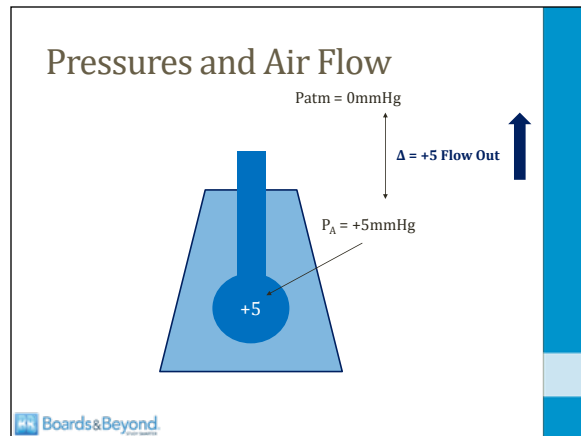


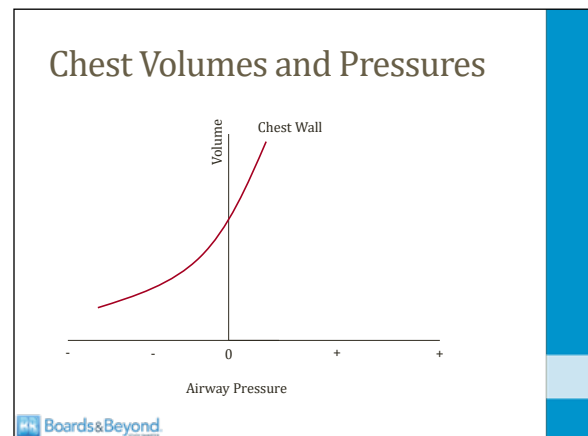
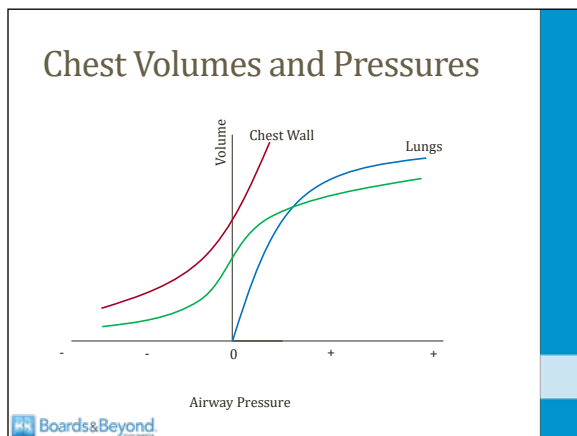
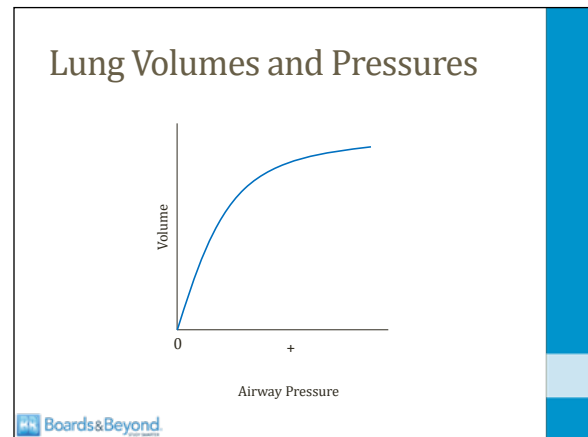
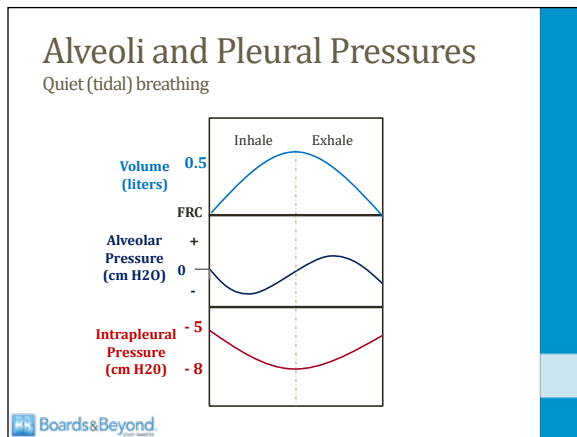
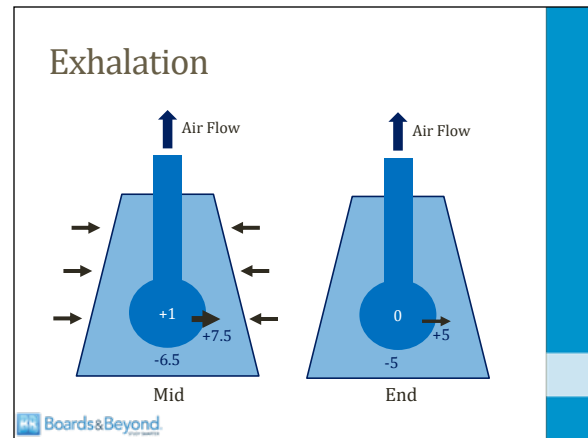
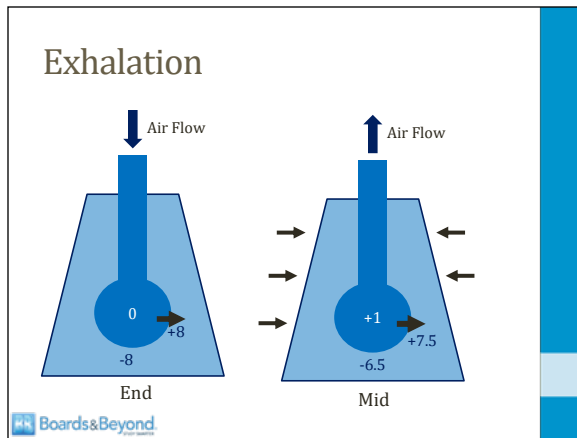
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## Pressures and Air Flow



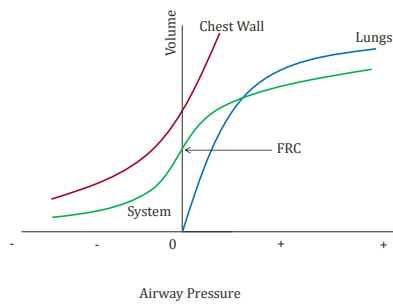
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## Chest Volumes and Pressures



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## Functional residual capacity

- Lung in = chest out
- Volume where lungs rest after quiet exhalation
- Pressure inside system is zero
  - No  $\uparrow$ / $\downarrow$  pressure from push/pull of lungs or chest wall
  - Pressure = atmospheric pressure

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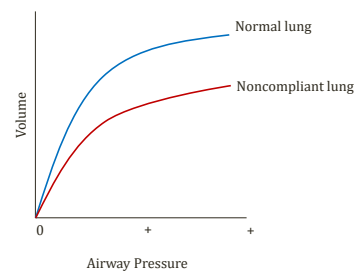
## Lung Compliance

- For given pressure how much volume changes
- Compliant lung
  - Small amount of diaphragm effort
  - Generates small pressure change across lungs
  - Large volume change
  - Easy to move air in/out
- Non-compliant lung
  - Large amount diaphragm effort
  - Big pressure change across lung
  - Small volume change (lungs stiff)
  - Hard to move air in/out

$$C = \frac{\Delta V}{\Delta P}$$

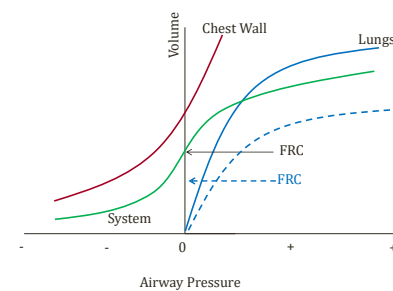
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## Lung Compliance



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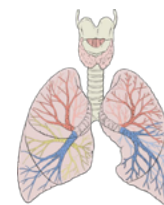
## Chest Volumes and Pressures



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## Lung Compliance

- **Decreased ( $\downarrow$  FRC)**
  - Pneumonia
  - Pulmonary edema
  - Pulmonary fibrosis
- **Increased ( $\uparrow$  FRC)**
  - Emphysema (floppy lungs)
  - Aging
  - Surfactant

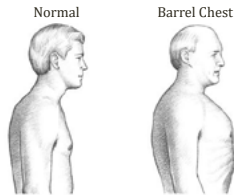


Patrick Lynch/Wikipedia

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## Barrel Chest

- Seen in patients with **emphysema**
- Increased lung compliance
- Increased FRC → larger volumes in chest

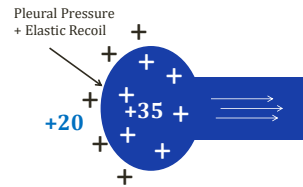


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Quora/Public Domain

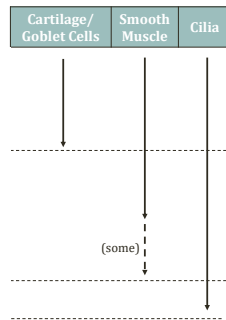
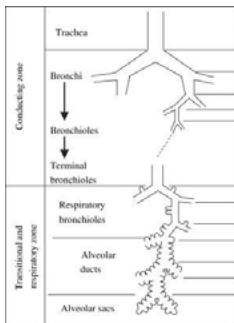
## Forced Exhalation

- Pleural pressure becomes **positive**
- Compresses airway
- Pressure on alveoli → positive pressure in airway
- Pushes air out → air flows from airways



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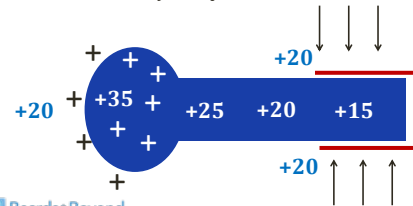
## Zones



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## Equal Pressure Point

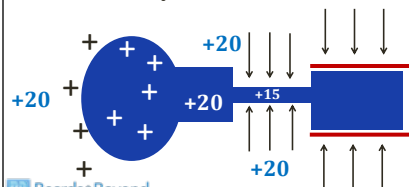
- Pleural pressure = airway pressure
- Beyond this point airway collapses
- In healthy lungs: EPP occurs in **cartilaginous** airways
- Prevents airway collapse



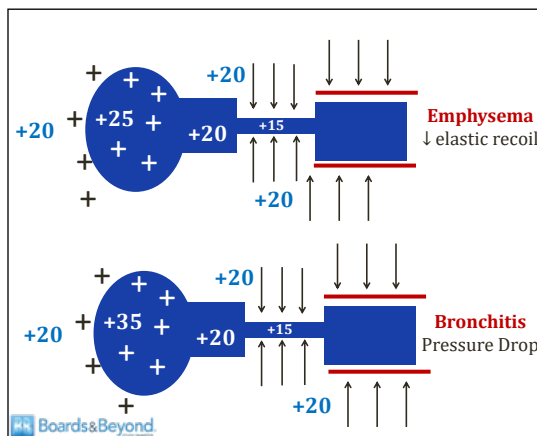
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## Equal Pressure Point

- In disease: EPP **moves toward alveoli**
  - Obstruction (bronchitis): more pressure drop
  - Emphysema: loss of elastic recoil
- Can be reached in thin-walled bronchioles
- Result: Collapse, obstruction to airflow, air trapping



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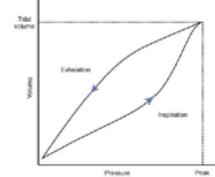
## COPD

Chronic Obstructive Pulmonary Disease

- **Slow exhalation**
  - Prevents large rise in pleural pressure
  - Forceful exhalation would  $\uparrow\uparrow$  intrapleural pressure
- **Pursed lips**
  - Increases airway (alveolar) pressure
  - Prevents collapse

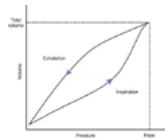
## Hysteresis

- Hysteresis = dependence of property on its history
- Different PV curves for inhalation and exhalation
- Slope PV curve = compliance
- Different compliance despite same lung structures



## Hysteresis

- PV hysteresis caused by **surface tension**
- Inspiration begins with smallest volume
  - Molecules close together
  - **Strongest surface tension**
- Expiration begins at high lung volumes
  - Intermolecular forces low



# Hemoglobin

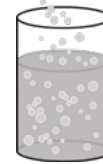
## Hemoglobin

Jason Ryan, MD, MPH

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## Oxygen Transport

- **Dissolved  $O_2$** 
  - Determined by Henry's law
  - $P_{aO_2} \times \text{solubility} = \text{dissolved } O_2$
  - Very small amount (2%) total blood  $O_2$
- **Bound to hemoglobin (98%)**

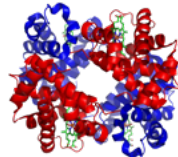


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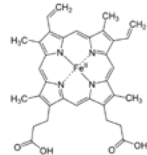
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## Hemoglobin

- Globin chains
  - Proteins
  - Alpha ( $\alpha$ )
  - Beta ( $\beta$ )
  - Gamma ( $\gamma$ )
  - Delta ( $\delta$ )
  - 4 chains in 2 pairs
- Heme
  - Molecule (non-peptide)
  - Contains iron (Fe)
  - Porphyrin ring
  - Oxygen binds iron



Richard Wender and Zephryn



Boards&Beyond

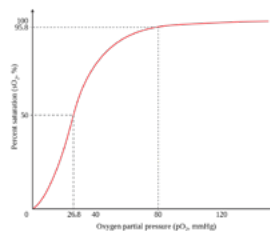
## Hemoglobin Types

- **Hemoglobin A**
  - Adult type
  - Most common type found (95%)
  - $\alpha_2 \beta_2$
- **Hemoglobin A2**
  - Adult type
  - Less common type (2-3%)
  - $\alpha_2 \delta_2$
- **Hemoglobin F**
  - Fetal type
  - $\alpha_2 \gamma_2$

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## O<sub>2</sub>-Hgb Dissociation Curves

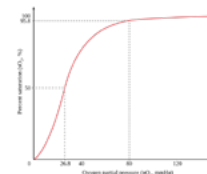
- Y axis: percentage of hemoglobin bound to oxygen
- X-axis: partial pressure of oxygen ( $P_{aO_2}$ )



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## Oxygen-Hgb Binding

- Four heme groups do not simultaneously oxygenate
- First  $O_2$  molecule INCREASES affinity for 2<sup>nd</sup> molecule
  - Second  $O_2$  molecule INCREASES affinity for 3<sup>rd</sup> molecule
  - Third  $O_2$  molecule INCREASES affinity for 4<sup>th</sup> molecule
  - Affinity last  $O_2 = 300 \text{ times}$  affinity for first  $O_2$
- **Positive cooperativity**
- Makes curve S shaped

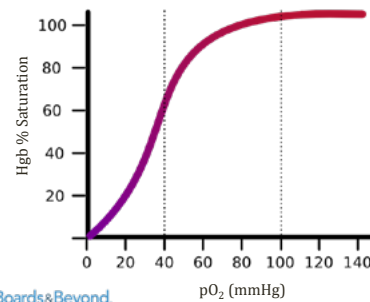


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## Allosteric Proteins

- Allosteric = "other site"
- Binding at one site influenced by other sites
- Usually multi-subunit proteins
- Hemoglobin is an allosteric structure
- O<sub>2</sub> cooperativity is a **positive** allosteric effect

## O<sub>2</sub>-Hgb Dissociation Curves

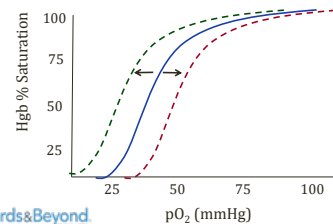


## Hemoglobin Forms

- Globin chains can assume two formations
- **Taut form (T)**
  - Tends to release O<sub>2</sub>
  - Favored form in tissues
  - Allows more release of O<sub>2</sub>
- **Relaxed form (R)**
  - Holds on to O<sub>2</sub>
  - Favored form in lungs
  - Allows more binding of O<sub>2</sub>

## Shifts in O<sub>2</sub>-Hgb Curves

- Affinity of Hgb for O<sub>2</sub> can change – not fixed
- Hgb modified by environment within RBCs
- Dissociation curve shifts may occur to right or left



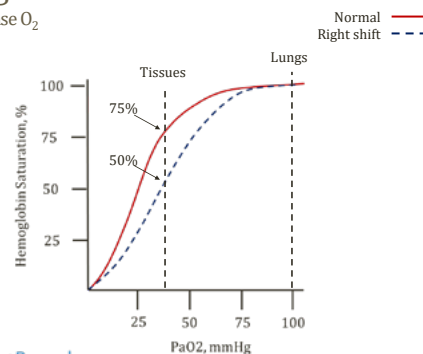
## Rightward Shift

|             | Lungs PaO <sub>2</sub> | Lungs %Sat | Tissues PaO <sub>2</sub> | Tissues %Sat |
|-------------|------------------------|------------|--------------------------|--------------|
| Normal      | 100                    | 100        | 40                       | <b>75%</b>   |
| Right Shift | 100                    | 100        | 40                       | <b>50%</b>   |

Normal: 100% → 75% **Δ 25%**  
 Right shift: 100% → 50% **Δ 50%**

## Right Curve Shifts

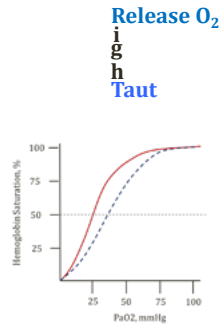
Release O<sub>2</sub>



## Right Curve Shifts

Release O<sub>2</sub>

- Favors **taut form**
- Causes of **right shifts**
  - Rising Metabolic Activity
  - ↑CO<sub>2</sub>
  - ↓pH
  - ↑Temp
- Increases P50

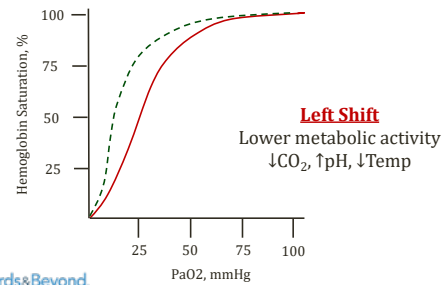


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## Left Curve Shifts

Latch on to O<sub>2</sub>

Normal —  
Left shift - - -

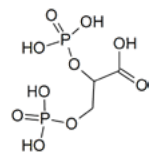


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## 2,3-Bisphosphoglycerate

2,3 BPG

- Found in RBCs
- Promotes O<sub>2</sub> release from hemoglobin
- *Negative* allosteric effector
- Increasing levels:
  - Decrease oxygen affinity of hemoglobin
  - Increase delivery oxygen to tissues

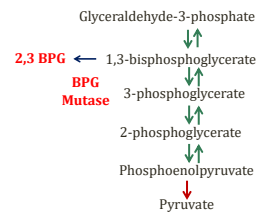


2,3-Bisphosphoglycerate

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## 2,3 Bisphosphoglycerate

- Created from diverted 1,3 BPG (glycolysis)
- **Sacrifices ATP** from glycolysis

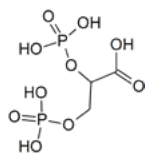


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## 2,3-Bisphosphoglycerate

2,3 BPG

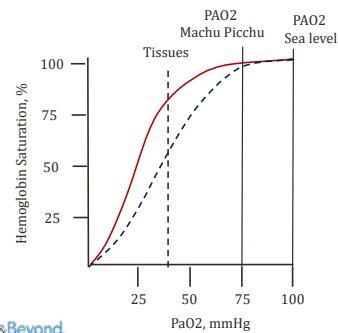
- ↑ BPG with **chronic hypoxia**
  - COPD
  - High altitude
  - Chronic anemia



2,3-Bisphosphoglycerate

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## High Altitude



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## Fetal Hemoglobin

HgbF ( $\alpha_2\gamma_2$ )

- After 8 weeks HgbF is predominant Hgb
  - Up to 90% fetal hemoglobin
  - Levels fall in weeks/months after birth
  - In adult HgbF <1% total hemoglobin
- Higher  $O_2$  affinity than HgbA
  - Necessary because fetal  $pO_2 = 40\text{mmHg}$

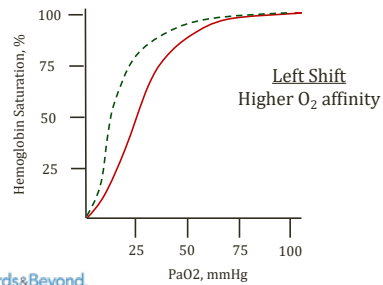


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## Fetal Hemoglobin

HgbF ( $\alpha_2\gamma_2$ )

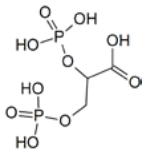


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## Fetal Hemoglobin

HgbF ( $\alpha_2\gamma_2$ )

- Left shift caused by altered **2,3 BPG binding**
  - 2,3-BPG binds  $\gamma$  chains poorly (binds  $\beta$  chains avidly)
  - Less 2,3-BPG binding  $\rightarrow O_2$  affinity increases (left shift)

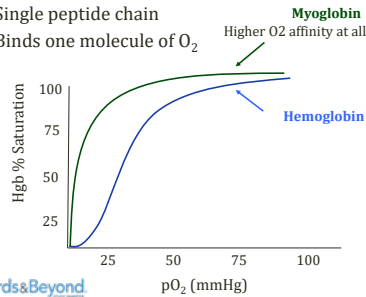


2,3-Bisphosphoglycerate

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## Myoglobin

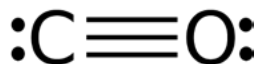
- Found in skeletal muscle and heart
- Single peptide chain
- Binds one molecule of  $O_2$



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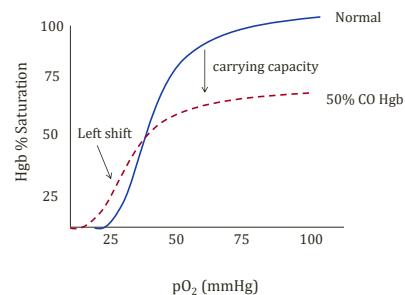
## Carbon Monoxide

- Binds to iron in heme 240x the affinity of  $O_2$
- Forms **carboxyhemoglobin** (HbCO)
- Blocks  $O_2$  binding sites (less  $O_2$  can be absorbed)
- "Functional anemia"
- Other binding sites cannot offload  $O_2$ 
  - Allosteric modification of hemoglobin
  - Shifts dissociation curve left



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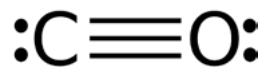
## Carbon Monoxide



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## Carbon Monoxide Poisoning

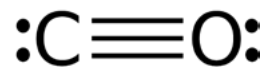
- Nonspecific symptoms
- Headache most common
- Malaise, nausea, dizziness
- Classic (but rare) sign: **cherry red lips**
  - Carboxyhemoglobin is red
  - Do not see blue lips (cyanosis)



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## Carbon Monoxide Poisoning

- Standard pulse oximetry normal
  - Cannot differentiate carboxyhemoglobin/oxyhemoglobin
- Diagnosis: **carboxyhemoglobin level**
  - Normal <3%
  - Smokers 10-15%
  - >15% suggest poisoning
- Treatment: **oxygen**



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## Methemoglobinemia

- Most iron in hemoglobin normally reduced ( $\text{Fe}^{2+}$ )
- Small amount oxidizes iron:  $\text{Fe}^{3+}$ 
  - Called **methemoglobin**
  - Cannot bind  $\text{O}_2$
- Excess methemoglobin: hypoxia



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## Methemoglobinemia

- Acquired methemoglobinemia from **drugs**
  - Local anesthetics (benzocaine)
  - Nitric oxide
  - Dapsone
- Treatment: **methylene blue**
  - Reducing agent
  - $\text{Fe}^{3+} \rightarrow \text{Fe}^{2+}$

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## Clinical Scenario

- Endoscopy patient
- **Benzocaine** spray used for throat analgesia
- Post procedure shortness of breath
- "Chocolate brown blood"
- $\text{O}_2$  sat (pulse oximetry) = variable (80s-90s)
- Oxygen does not improve shortness of breath
- $\text{P}_{\text{aO}_2}$  (blood gas) = normal
- Diagnosis:  $\uparrow$  methemoglobin level
- Other example:
  - Premature babies given **NO** for pulmonary vasodilation

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# Pulmonary Circulation

## Pulmonary Circulation

Jason Ryan, MD, MPH

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## Pulmonary Circulation

- Low pressure system
  - Systemic: 120/80
  - Pulmonary artery: 24/12
- Walls of pulmonary artery very thin
  - Little smooth muscle
  - Low resistance to flow
  - Very distensible (compliant)

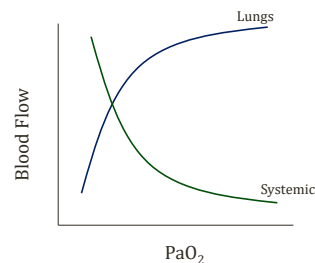
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## Blood Oxygen Content

- Systemic circulation
  - ↓  $O_2$  level ( $PaO_2$ ) leads to vasodilation (↑ blood flow)
- Pulmonary circulation
  - ↓  $O_2$  level ( $PaO_2$ ) leads to **vasoconstriction** (↓ blood flow)
  - "Hypoxic vasoconstriction"
  - Shunts blood away from poorly ventilated areas
  - More blood to well ventilated areas
- Key for fetal circulation
  - Low  $O_2$  constricts pulmonary arteries in womb
  - At birth, arteries receive  $O_2$  and dilate

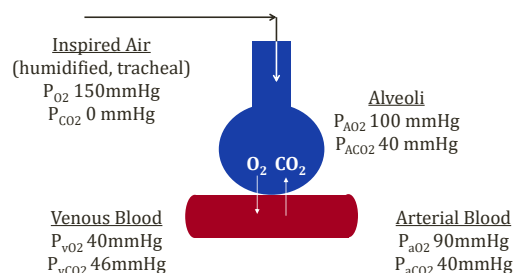
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## Blood $O_2$ Content



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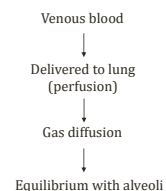
## Gas Exchange



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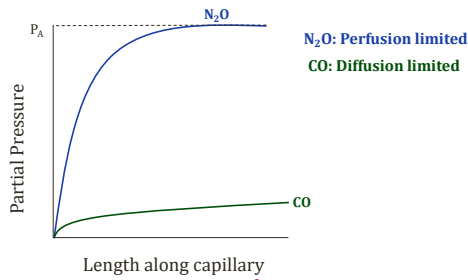
## Gas Exchange

- Gasses classified by limiting factor for gas transfer
- **Perfusion limited**
  - Gas transport limited by perfusion (blood flow)
  - More blood flow → more uptake of gas
- **Diffusion limited**
  - Gas transport limited by *diffusion*



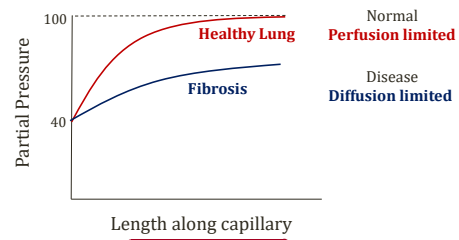
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## Gas Exchange



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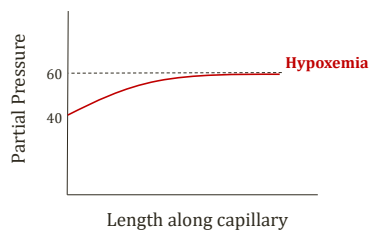
## Gas Exchange: Oxygen



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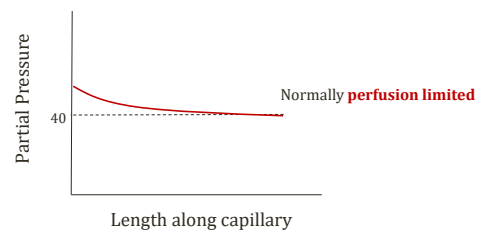
## Gas Exchange: Oxygen

High Altitude



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## Gas Exchange: Carbon Dioxide



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## DLCO

Diffusing capacity of carbon monoxide

- Measures ability of lungs to transfer gas
- Patient inhales small amount (not dangerous) CO
- CO uptake is **diffusion limited**
  - Amount taken up  $\approx$  diffusion capacity of lungs
- Machine measures CO exhaled
- Normal = 75 - 140 % predicted
- Severe disease <40% predicted

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## Low DLCO Disorders

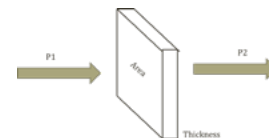
### • Emphysema

- Destruction of alveoli
- Decreased surface area

### • Fibrosis or pulmonary edema

- Diffusion distance (thickness) increases

$$V_{\text{gas}} = \frac{\text{Area} \cdot D \cdot (P_1 - P_2)}{\text{Thickness}}$$



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## Resistance to Blood Flow

### Pulmonary Vascular Resistance

- Two vessels types:
  - Alveolar: capillaries
  - Extra-alveolar: arteries and veins
- Increased lung volumes:
  - Crushes alveolar vessels → high resistance
  - Pulls extra-alveolar vessels open

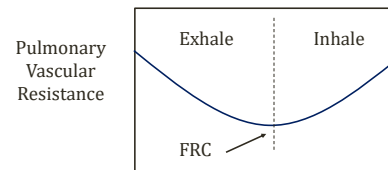
Artery

Capillary

Vein

## Pulmonary Vascular Resistance

### Resistance to blood flow



## Pulmonary Hypertension

- Normal PA pressure
  - 24/12
  - Mean 10-14mmHg
- **Pulmonary hypertension**
  - Mean pressure >25mmHg
- **Loud P2** = pulmonary hypertension
  - "Accentuated" or "loud" second heart sound
  - Left upper sternal border



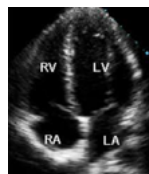
## Pulmonary Hypertension

- Main symptom is dyspnea
- Untreated can lead to **"cor pulmonale"**
  - Chronic high pressure in right ventricle
  - Right ventricle hypertrophies
  - Eventually dilates and fails
  - Jugular venous distension
  - Lower extremity edema
  - Hepatomegaly
- Death from heart failure or arrhythmia



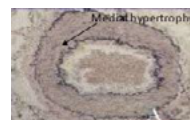
## Pulmonary Hypertension

- Gold standard diagnosis: **right heart catheterization**
- Non-invasive diagnosis by **echocardiography**
  - Estimate PA pressure
  - Visualize right heart structures

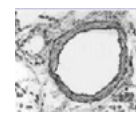


## Pulmonary Hypertension

- **Arteriosclerosis**
  - Thickening of arterial walls
- Proliferation smooth muscle cells
  - Thickening media
  - Narrowing of the lumen

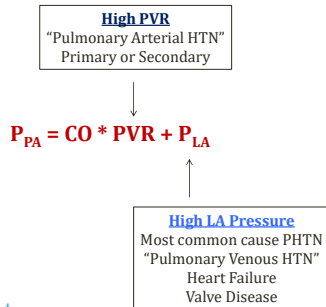


Medial Hypertrophy



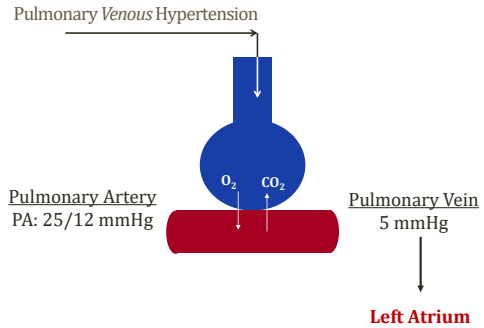
Normal

## Pulmonary Hypertension



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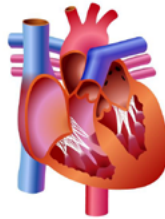
## Left Heart Disease



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## Left Heart Disease

- Most common cause of pulmonary hypertension
- "Pulmonary venous hypertension"
- Any cause of **high left atrial pressure**
  - Heart failure
  - Mitral stenosis
  - Mitral regurgitation



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## High PVR

Pulmonary Arterial Hypertension

- **Hypoxemia** → vasoconstriction
  - COPD, other chronic lung diseases
  - Sleep apnea or high altitude (chronic hypoxia)
- **Chronic pulmonary emboli**
  - Decreased **area** for blood flow



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## PAH

Pulmonary Arterial Hypertension

- High pulmonary vascular resistance
- No chronic lung disease or thrombosis
- Key associations:
  - **Connective tissue disease (scleroderma)**
  - Human immunodeficiency virus
  - Congenital heart disease (shunts)
  - Schistosomiasis
  - Drugs (amphetamines, cocaine)

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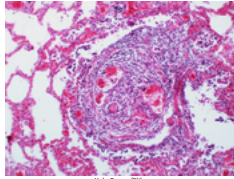
## Idiopathic PAH

- Rare disease
- Classically affects young women
- **High pulmonary vascular resistance**
- Increased activity vasoconstrictors
  - Endothelin
- Decreased activity vasodilators
  - Nitric oxide

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## Plexiform Lesions

- Unique to idiopathic PAH
- Endothelial proliferation forms **multiple lumens**
- Small arteries branch points from medium arteries



Yale Rosen/PHH

## BMPR2 gene mutations

- Bone morphogenetic protein receptor type II
  - Inhibits smooth muscle proliferation
  - Mutations → abnormal growth (endothelium, smooth muscle)
- Up to 25% of idiopathic cases
- Up to 80% **familial cases**

## PAH Treatments

- All lower PVR
- Epoprostenol: Prostacyclin (IV)
  - $\text{PGI}_2$
  - Potent vasodilator
- Bosentan:
  - Antagonist endothelin-1 receptors (PO)
- Sildenafil:
  - Inhibits PDE-5 in smooth muscle of lungs (PO)

# Ventilation & Perfusion

## Ventilation & Perfusion

Jason Ryan, MD, MPH

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## Ventilation

- Ventilation = volume x frequency (respiratory rate)
  - 500cc per breath x 20 breaths per minute
  - 10,000cc/min
- **Alveolar ventilation** = useful for gas exchange
- **Dead space ventilation** = wasted ventilation



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## Dead Space

- Filled with air but no gas exchange
- **Anatomic dead space**
  - Volume of conducting portions of respiratory tract
  - Nose, trachea
- **Physiologic dead space**
  - Anatomic PLUS volume of alveoli that don't exchange gas
  - Includes functional dead space
  - Insufficient perfusion
  - Apex is largest contributor
- Physiologic dead space increases many diseases

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## Measuring Dead Space

- **Bohr's method**
- Physiologic dead space ( $V_d$ ) from:
  - Tidal volume ( $V_t$ )
  - $P_{eCO_2}$  (exhaled air)
  - $P_{aCO_2}$  (blood gas)

$$\frac{V_d}{V_t} = \frac{P_{aCO_2} - P_{eCO_2}}{P_{aCO_2}}$$

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## Nomenclature

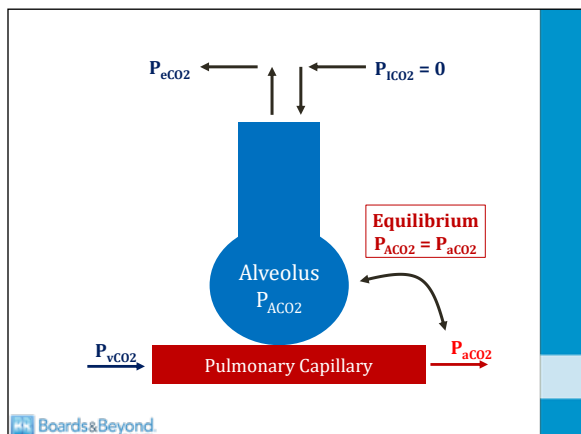
- $P_A$  = alveolar pressure
  - $P_{AO_2}$  = alveolar  $O_2$
  - $P_{ACO_2}$  = alveolar  $CO_2$
- $P_a$  = arterial pressure
  - $P_{aO_2}$  = arterial  $O_2$
  - $P_{aCO_2}$  = arterial  $CO_2$
- $P_v$  = venous pressure
- $P_e$  = expired pressure

**A = alveolar  
a = arterial**

LEARNING TO WRITE THE LETTER A



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## Bohr Equation

$$\frac{V_d}{V_t} = \frac{P_a\text{CO}_2 - P_e\text{CO}_2}{P_a\text{CO}_2}$$

## Zero Dead Space

$$\frac{V_d}{V_t} = \frac{P_a\text{CO}_2 - P_e\text{CO}_2}{P_a\text{CO}_2}$$

$$0 = \frac{P_a\text{CO}_2 - P_e\text{CO}_2}{P_a\text{CO}_2}$$

$$0 = P_a\text{CO}_2 - P_e\text{CO}_2$$

$$P_e\text{CO}_2 = P_a\text{CO}_2$$

↓ dead space →  $P_e\text{CO}_2$  approaches  $P_a\text{CO}_2$   
More gas exchange  
Less retained  $\text{CO}_2$

## 100% Dead Space

$$\frac{V_d}{V_t} = \frac{P_a\text{CO}_2 - P_e\text{CO}_2}{P_a\text{CO}_2}$$

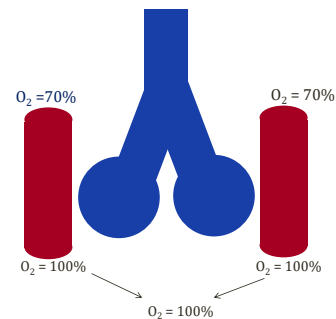
$$1 = \frac{P_a\text{CO}_2 - P_e\text{CO}_2}{P_a\text{CO}_2}$$

$$P_a\text{CO}_2 = P_a\text{CO}_2 - P_e\text{CO}_2$$

$$P_e\text{CO}_2 = 0$$

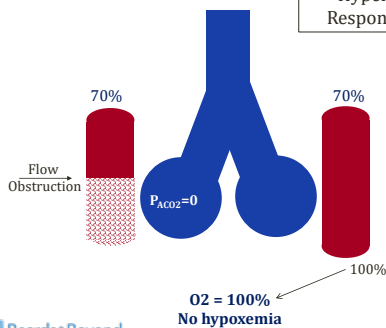
↑ dead space →  $P_e\text{CO}_2$  approaches zero  
Less gas exchange  
More retained  $\text{CO}_2$

## Dead Space



## Dead Space

Main problem: ↑  $\text{CO}_2$   
Hypercapnia  
Response: ↑ RR



## Alveolar Ventilation Equation

Predicts Alveolar  $\text{CO}_2$

- Total ventilation (TV) = volume/min
  - \*\*Volume in slightly > volume out due to  $\text{O}_2$  uptake
  - Sometimes called minute ventilation
- **Alveolar ventilation**
  - TV minus "dead space"
- Example: 500cc per minute
  - 150cc fills dead space
  - Only 350cc available for gas exchange



## Elevated Carbon Dioxide

- Hypercapnia
- Hypercarbia
- Causes acidosis
- Physiologic response: **↑ respiratory rate**
  - Increased alveolar ventilation

## Alveolar Ventilation Equation

Predicts Alveolar CO<sub>2</sub>

- V<sub>A</sub> = alveolar ventilation
- V<sub>CO<sub>2</sub></sub> = rate of CO<sub>2</sub> production
- P<sub>ACO<sub>2</sub></sub> = alveolar PCO<sub>2</sub>
- V<sub>T</sub> = total ventilation
- V<sub>ds</sub> = dead space ventilation
- K = constant

**Three Major Causes of ↑ CO<sub>2</sub>**

↑ CO<sub>2</sub> production  
↓ V<sub>A</sub> (hypoventilation)  
↑ V<sub>ds</sub> (dead space)

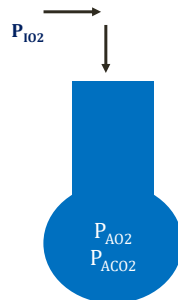
$$P_{ACO_2} = \frac{V_{CO_2} * K}{V_A} \quad P_{ACO_2} = \frac{V_{CO_2} * K}{V_T - V_{ds}}$$

## Alveolar Gas Equation

Predicts Alveolar O<sub>2</sub>

- P<sub>AO<sub>2</sub></sub> = alveolar O<sub>2</sub>
- P<sub>IO<sub>2</sub></sub> = inspired O<sub>2</sub>
- P<sub>ACO<sub>2</sub></sub> = alveolar CO<sub>2</sub>
- R = respiratory exchange ratio
  - CO<sub>2</sub> production/O<sub>2</sub> consumption
  - Varies with diet, metabolic state

$$P_{AO_2} = P_{IO_2} - \frac{P_{ACO_2}}{R}$$



## Alveolar Gas Equation

Predicts Alveolar O<sub>2</sub>

| P <sub>ACO<sub>2</sub></sub> | P <sub>IO<sub>2</sub></sub> | P <sub>AO<sub>2</sub></sub> |
|------------------------------|-----------------------------|-----------------------------|
| 40                           | 150                         | 100                         |
| 50                           | 150                         | 88                          |
| 60                           | 150                         | 75                          |
| 70                           | 150                         | 63                          |
| 80                           | 150                         | 50                          |

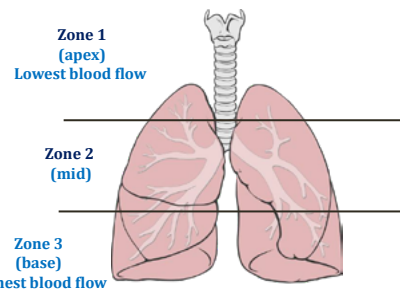
$$P_{AO_2} = P_{IO_2} - \frac{P_{ACO_2}}{R}$$

## Lung Perfusion

- Upright position: Blood flow distribution is **uneven**
  - Caused by **gravity**
  - Apex: Lowest blood flow
  - Base: Highest blood flow
- Lung divided into **3 zones** to describe perfusion



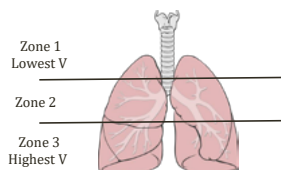
## Lung Perfusion





## Lung Ventilation

- Ventilation highest zone 3, lowest zone 1
  - Also caused by gravity
  - Upper lung compresses base → pushes air out
  - More room for filling of base with next breath
  - Variations smaller (L/min) than blood flow



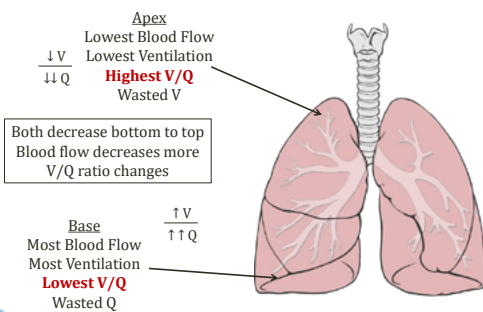
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## Ventilation-Perfusion Ratio

- V/Q ratio: alveolar ventilation/pulmonary blood flow
  - Matching critical for gas exchange
  - Under-ventilated or under-perfused alveoli inefficient
- **Normal V/Q ratio = 0.8**
  - Alveolar ventilation (L/min)/pulmonary blood flow (L/min)
  - Yields normal  $P_{aO_2}$  (90 mmHg) and  $P_{aCO_2}$  (40 mm Hg)

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## Ventilation-Perfusion Ratio



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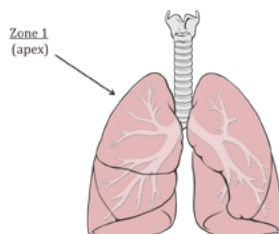
## Ventilation-Perfusion Ratio

|        | V | Q | V/Q        | $P_{aO_2}$ | $P_{aCO_2}$ |
|--------|---|---|------------|------------|-------------|
| Zone 1 | ↓ | ↓ | ↑<br>(3.0) | 130        | 30          |
| Zone 2 |   |   |            |            |             |
| Zone 3 | ↑ | ↑ | ↓<br>(0.6) | 90         | 42          |

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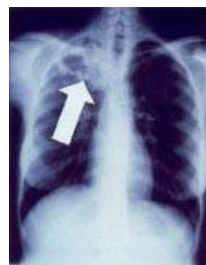
## Zone 1

- Lowest blood flow
- Lowest ventilation
- Highest V/Q ratio
- Highest  $P_{aO_2}$
- Lowest  $P_{aCO_2}$



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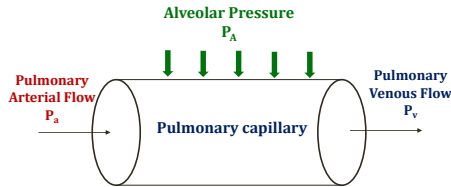
## Tuberculosis



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## Pulmonary Blood Flow

- Normally, A-V pressure difference drives blood flow
- In lungs, alveolar pressure may determine blood flow
- High alveolar pressure  $\rightarrow$  no blood flow  $\rightarrow$  dead space

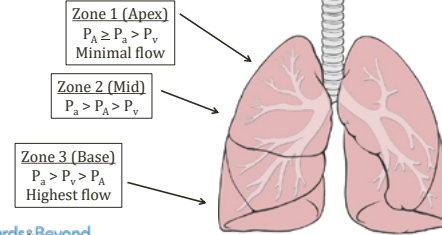


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## Pulmonary Blood Flow

**Pressures**  
 $P_A$  Alveolar  
 $P_a$  Arterial  
 $P_v$  Venous

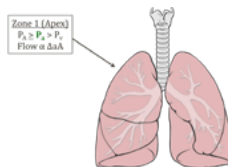
- $P_A$  constant
- At base,  $P_a$  and  $P_v$  highest
- At apex,  $P_a$  and  $P_v$  lowest



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## Zone 1

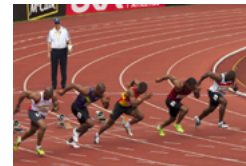
- Lung apex:  $P_A \geq P_a > P_v$
- Slight fall in  $P_a \rightarrow$  **capillary compression**
  - Hemorrhage/shock
- Zone 1 becomes **dead space**
  - Ventilation without perfusion



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## Exercise

- Increased  $O_2$  demand
- Ventilation rate increases
- Increased cardiac output
- **V/Q ratio approaches 1**
  - More blood flow
  - More ventilation
  - $\uparrow$  ventilation  $>$   $\uparrow$  blood flow
  - Becomes more even in zones

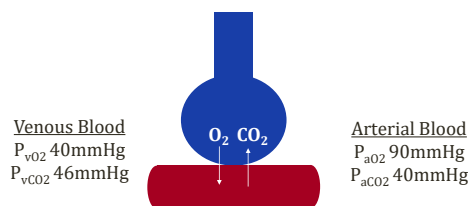


William Warby/Flickr

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## Exercise

- No change in mean  $P_{aO_2}$  and  $P_{aCO_2}$
- Increased venous  $CO_2$  ( $P_{vCO_2}$ )
- Decreased venous  $O_2$  ( $P_{vO_2}$ )



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# Hypoxia

## Hypoxia

Jason Ryan, MD, MPH

Boards&Beyond

## Oxygen delivery to tissues

- Oxygen delivery to tissues depends on:
  - Cardiac output
  - O<sub>2</sub> content of blood
- For proper O<sub>2</sub> delivery need:
  - Normal cardiac output
  - Normal O<sub>2</sub> content

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## What determines O<sub>2</sub> content?

- O<sub>2</sub> binding capacity
  - How much O<sub>2</sub> blood can hold
  - Determined by hemoglobin
- Hemoglobin saturation
  - % Hemoglobin molecules saturated
- Dissolved O<sub>2</sub>
  - O<sub>2</sub> directly dissolved in blood

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## PaO<sub>2</sub>

- Partial pressure oxygen in blood
- Obtained from an arterial blood gas
- Reflects amount of O<sub>2</sub> dissolved in blood
- Normal: >80mmHg



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## Pulse Oximetry

- Measures Hgb-O<sub>2</sub> saturation of blood
- Related to PaO<sub>2</sub>
- Uses light and a photodetector

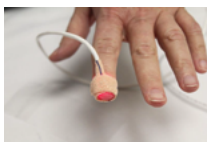
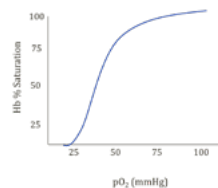


Image courtesy of Quinn Dombrowski



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## Oxygen Content

$$\text{O}_2 \text{ Content (ml O}_2\text{/dl)} = (\text{O}_2 \text{ Binding Capacity (1.39 * Hgb)}) * (\% \text{ Sat}) + (\text{Dissolved O}_2 \text{ (0.003 P}_{a\text{O}_2}))$$

- **Normal O<sub>2</sub> content requires:**
  - Presence of hemoglobin
  - Sufficient saturation of hemoglobin
  - Normal P<sub>aO2</sub>

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## Hypoxemia, Hypoxia, Ischemia

- Hypoxemia: low oxygen content of blood
- Hypoxia: low  $O_2$  delivery to tissues
- Ischemia: loss of blood flow



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## Hypoxemia, Hypoxia, Ischemia

- Low Hgb- $O_2$  sat or low  $PaO_2$  = hypoxemia
- Hypoxemia → hypoxia
- Can have hypoxia **without hypoxemia**

### Common Hypoxia Causes

Hypoxemia  
Heart Failure  
Anemia  
Carbon Monoxide

## Heart Failure

- ↓ **cardiac output**
- ↓ blood flow to tissues → hypoxia
- $O_2$  content of blood may be normal
- $PaO_2$  and Hgb- $O_2$  sat may be normal



## Anemia

- Oxygenation of blood by lungs is normal
- **Oxygen carrying capacity** of blood reduced
- Low  $O_2$  content of blood
- $PaO_2$  and Hgb- $O_2$  sat normal



Database Center for Life Science (DBCLS)

## Carbon Monoxide

- Binds to iron in heme 240x the affinity of oxygen
- Blocks  $O_2$  binding sites: **"functional anemia"**
- Alveolar  $O_2$  ( $P_{A02}$ ) usually normal
  - Amount of CO gas required for poisoning usually small
- Normal  $P_{A02}$  → Normal  $P_{a02}$ 
  - ↓  $O_2$  binding to Hb despite normal  $P_{a02}$

## Carbon Monoxide

- Low Hgb- $O_2$  sat (CO blocking  $O_2$  binding sites)
- Pulse oximeter shows normal (100%)  $O_2$  sat
  - Can't distinguish Hb bound to CO from that bound to  $O_2$
- $O_2$  content of blood reduced

**Normal  $PaO_2$**   
**Low  $O_2$  % sat (reality)**  
**Normal  $O_2$  % sat (detector)**  
**Hypoxia**

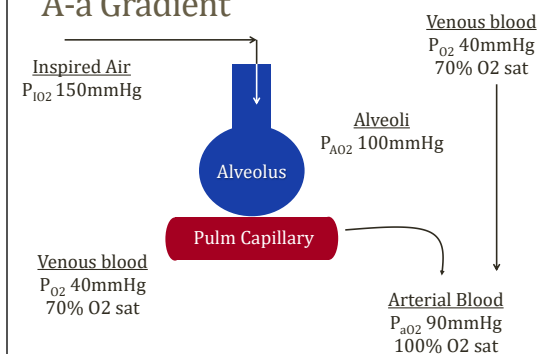
## Causes of Hypoxia

|                 | O2 Content | PaO2   | % Sat  |
|-----------------|------------|--------|--------|
| Hypoxemia       | ↓          | ↓      | ↓      |
| Heart Failure   | Normal     | Normal | Normal |
| Anemia          | ↓          | Normal | Normal |
| Carbon Monoxide | ↓          | Normal | ↓*     |

## Hypoxemia

- Indicates defect oxygenating blood
- Causes categorized by **A-a gradient**
  - Alveolar O<sub>2</sub> (P<sub>A02</sub>) - Arterial O<sub>2</sub> (P<sub>a02</sub>)
  - P<sub>A02</sub> from alveolar gas equation
  - P<sub>a02</sub> from blood gas

## A-a Gradient



## Alveolar Gas Equation

$$P_{A02} = P_{I02} - \frac{P_{aCO2}}{R} = 150 - \frac{P_{aCO2}}{0.8}$$

## A-a Gradient

- Difference between alveolar (A) and arterial (a) O<sub>2</sub>
- Helpful for evaluating hypoxemia
- Step 1: Measure P<sub>a02</sub>, P<sub>aCO2</sub>
- Step 2: Determine P<sub>A02</sub> from gas equation
- Step 3: A-a gradient = P<sub>A02</sub> - P<sub>a02</sub>
- Normal 10-15mmHg
  - Shunting from **thebesian** and **bronchial veins**

## Normal A-a Gradient

- **Low alveolar oxygen content (P<sub>A02</sub>)**
- **Decreased oxygen content of air**
  - High altitude
  - P<sub>I02</sub> sea level = 150 mmHg
  - P<sub>I02</sub> high altitude ~ 100 mmHg
- **Hypoventilation**
  - Reduced respiratory rate
  - Reduced tidal volume
  - Causes increase P<sub>aCO2</sub> → decreased P<sub>A02</sub>
  - Narcotics, neuromuscular weakness, obesity

## Normal A-a Gradient

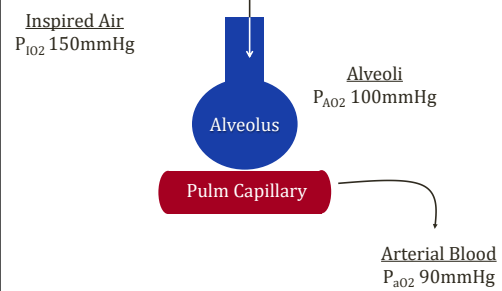
- Improves with oxygen

$$P_{A_{O_2}} = P_{I_{O_2}} - \frac{P_{a_{CO_2}}}{R} = 150 - \frac{P_{a_{CO_2}}}{0.8}$$



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## A-a Gradient



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## Alveolar Gas Equation

Hypoventilation  
High  $CO_2$

↓

$$P_{A_{O_2}} = P_{I_{O_2}} - \frac{P_{a_{CO_2}}}{R} = 150 - \frac{P_{a_{CO_2}}}{0.8}$$

↑

Low  $O_2$   
Inspired Air

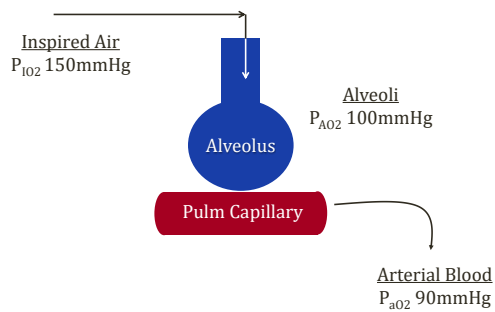
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## Increased A-a Gradient

- No problem with alveolar oxygen content ( $P_{A_{O_2}}$ )
- Low arterial oxygen content ( $P_{a_{O_2}}$ )**
- Most primary lung diseases: high A-a gradient
  - Pneumonia, pulmonary edema, etc.
- Three basic mechanisms create the high A-a gradient
  - Diffusion defects
  - Shunt
  - V/Q Mismatch

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## A-a Gradient



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## Diffusion

- Gases must diffuse from air to blood
- Rate of diffusion depends on:
  - Pressure difference (air-blood)
  - Area of alveoli for diffusion
  - Thickness of alveolar tissue

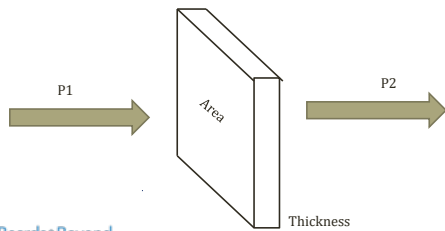
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## Diffusion

$$V_{\text{gas}} = \frac{\text{Area} * D * (P_1 - P_2)}{\text{Thickness}}$$

(vol/time)

(diff. coefficient)



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## Diffusion Limitation

$$V_{\text{gas}} = \frac{\text{Area} * D * (P_1 - P_2)}{\text{Thickness}}$$

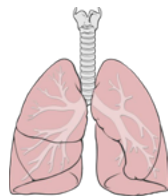
(vol/time)

- Surface area of alveoli falls in **emphysema**
- Diffusion distance (thickness) rises in:
  - **Pulmonary fibrosis**
  - **Pulmonary edema**
- Both lead to decreased diffusion → hypoxemia

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## Ventilation-Perfusion Ratio

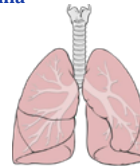
- V/Q ratio: alveolar ventilation/pulmonary blood flow
- Matching critical for gas exchange
- Unventilated or unperfused alveoli inefficient



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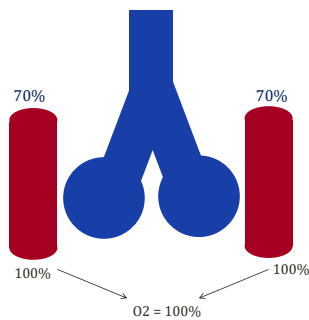
## Shunting

- No V
- Extreme reduction in V/Q
- **V/Q = 0**
- Venous blood to arterial system without oxygenation
- Causes **hypoxemia**



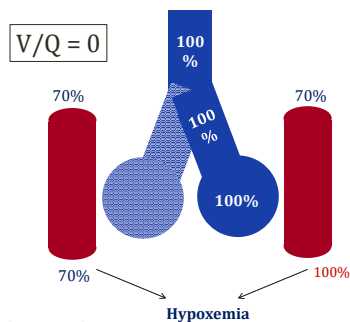
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## Shunting



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## Shunting

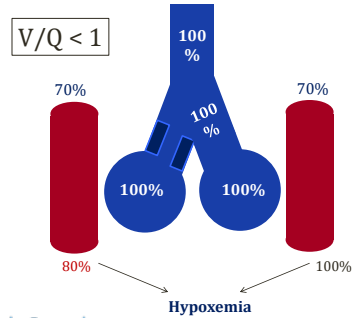


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## V/Q Mismatch

- $V/Q < 1$ 
  - **Reduced ventilation** relative to perfusion
  - Perfusion wasted
  - Blood going where not enough  $O_2$  present
  - Extreme version  $V/Q = 0$  is shunt
- Hypoxemia with increased A-a gradient
- **Improves with oxygen**

## V-Q Mismatch



## Carbon Dioxide

- Causes of hypercapnia
  - Hypoventilation
  - Increased dead space
  - Increased  $CO_2$  production
- Hypoxemia with high A-a gradient: **no  $\uparrow CO_2$**

## Mechanisms of Hypoxemia

High A-a gradient

|                      | PaO <sub>2</sub> | PaCO <sub>2</sub> | PaO <sub>2</sub> using 100% O <sub>2</sub> |
|----------------------|------------------|-------------------|--------------------------------------------|
| Diffusion Limitation | ↓                | --                | ↑                                          |
| Shunt<br>$V/Q = 0$   | ↓                | --                | No change                                  |
| $V/Q$ Mismatch       | ↓                | --                | ↑                                          |

## Mechanisms of Hypoxemia

|                     | PaO <sub>2</sub> | PAO <sub>2</sub> | A-a difference |
|---------------------|------------------|------------------|----------------|
| Normal A-a Gradient | ↓                | ↓                | 10-15          |
| High A-a Gradient   | ↓                | --               | Increased      |

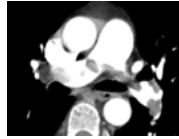
## Mechanisms by Disease

- Most diseases (COPD, PNA, pulm edema) have hypoxemia from multiple mechanisms
  - PNA may cause  $V/Q$  mismatch or shunt
- Some examples worth knowing
  - Intra-cardiac shunt: pure shunt mechanism
  - Inhale a peanut:  $V/Q = 0$  (also pure shunt)
  - Pulmonary Embolism



## Pulmonary Embolism

- Obstructed blood flow
- **↑ dead space**
- Hypoxemia does occur in many patients
- **V/Q mismatch**
  - Blood flow forced through open vessels
  - Increased Q (working vessels)
  - Same V
  - Decreased V/Q (mismatch)



# Carbon Dioxide

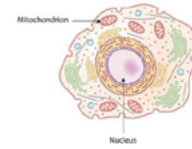
## Carbon Dioxide

Jason Ryan, MD, MPH

Boards&Beyond

## Carbon Dioxide

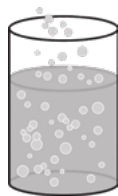
- Produced by **cellular metabolism**
- Transported to lungs via three mechanisms
  - Dissolved (5%)
  - Bound to hemoglobin (3%)
  - Bicarbonate (>90%)



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## Dissolved CO<sub>2</sub>

- Determined by **Henry's law**
- $\text{Pa}_{\text{CO}_2} \times \text{solubility} = \text{dissolved CO}_2$
- Very small amount (5%) total blood CO<sub>2</sub>



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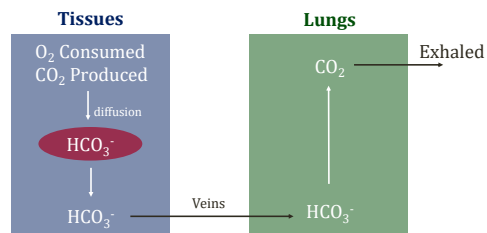
## Bicarbonate

- Most (>90%) CO<sub>2</sub> exists as bicarbonate
- Carrier form of CO<sub>2</sub>
- Red cells contain large amounts **carbonic anhydrase**
- Converts CO<sub>2</sub> to HCO<sub>3</sub><sup>-</sup>



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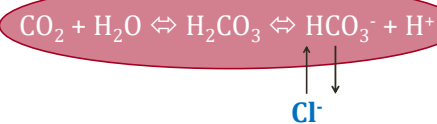
## Bicarbonate



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## RBC Bicarbonate Transport

- HCO<sub>3</sub><sup>-</sup> inside RBCs leaves cell to plasma
  - H<sup>+</sup> remains in red cells
- Chloride (Cl<sup>-</sup>)** enters cell
  - Maintains electrical neutrality
  - "Chloride shift"
- RBCs have high Cl<sup>-</sup> content in venous blood

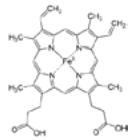
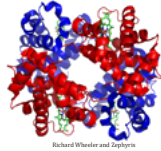


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## RBC Buffering H<sup>+</sup>

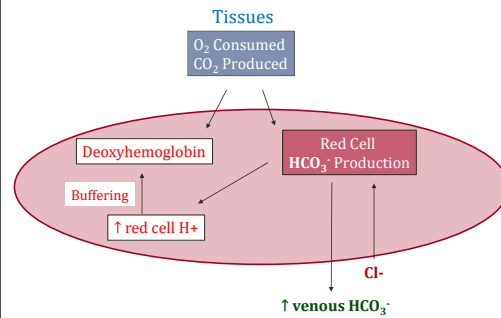


- H<sup>+</sup> produced when bicarbonate generated
  - Could cause dangerous fall in pH
- Deoxyhemoglobin** buffers (absorbs) H<sup>+</sup> in red cells
  - ↑ deoxyhemoglobin in RBCs when ↑ CO<sub>2</sub>
- Prevents H<sup>+</sup> from reducing pH



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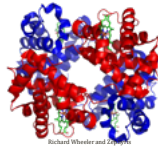
## Bicarbonate



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## Carbaminohemoglobin

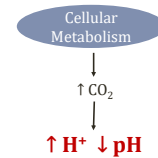
- Hemoglobin bound to CO<sub>2</sub>
  - Binds at different site from O<sub>2</sub>
- CO<sub>2</sub> binding alters affinity for oxygen
  - More CO<sub>2</sub> → More O<sub>2</sub> release
  - CO<sub>2</sub> decreases affinity for oxygen



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## Bohr Effect

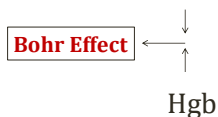
- CO<sub>2</sub> produced by metabolism → generates H<sup>+</sup> in RBCs
  - CO<sub>2</sub> + H<sub>2</sub>O ⇌ H<sub>2</sub>CO<sub>3</sub> ⇌ HCO<sub>3</sub><sup>-</sup> + H<sup>+</sup>
- H<sup>+</sup> and low pH are **indicators of metabolism**
- H<sup>+</sup> and low pH trigger **release of O<sub>2</sub>** by hemoglobin



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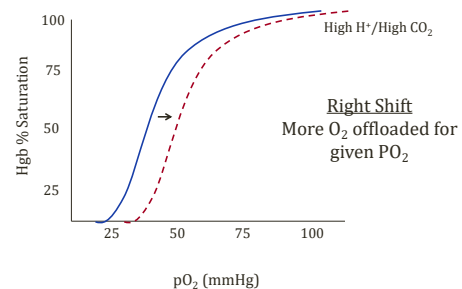
## Bohr Effect

- Deoxyhemoglobin has high affinity for H<sup>+</sup>
- H<sup>+</sup> binds hemoglobin in low O<sub>2</sub>/high CO<sub>2</sub> areas
- Converts Hgb to "taut form" which releases O<sub>2</sub>
  - Shifts O<sub>2</sub> curve to right
- Hemoglobin **releases more oxygen**



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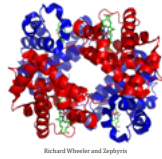
## Bohr Effect



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## Haldane Effect

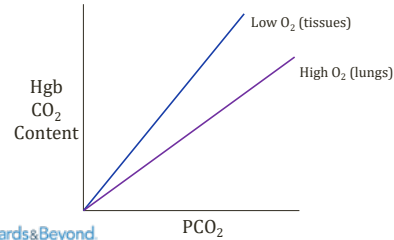
- O<sub>2</sub> binding alters **affinity for CO<sub>2</sub>**
  - Low O<sub>2</sub> environment Hgb binds more CO<sub>2</sub>
  - High O<sub>2</sub> environment Hgb binds *less* CO<sub>2</sub>



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## Haldane Effect

- Deoxyhemoglobin binds more CO<sub>2</sub>
  - Allows more CO<sub>2</sub> loading with O<sub>2</sub> consumption
  - Allows more CO<sub>2</sub> unloading with high O<sub>2</sub>



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## Tissues versus Lungs

### Tissues

- Low O<sub>2</sub> (consumption)
- High CO<sub>2</sub> (metabolism)
- High H<sup>+</sup>
- Low pH
- Favors O<sub>2</sub> unloading
  - Bohr Effect
- Favors CO<sub>2</sub> loading
  - Haldane Effect

### Lungs

- High O<sub>2</sub> (air)
- Low CO<sub>2</sub> (exhalation)
- Low H<sup>+</sup>
- High pH
- Favors O<sub>2</sub> loading
  - Bohr Effect
- Favors CO<sub>2</sub> unloading
  - Haldane Effect

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## CO<sub>2</sub> Transport

|                               | Lungs/Arteries | Tissues/Veins |
|-------------------------------|----------------|---------------|
| P <sub>O2</sub>               | 100            | ↓             |
| P <sub>CO2</sub>              | 40             | ↑             |
| HCO <sub>3</sub> <sup>-</sup> | 24             | ↑             |
| pH                            | 7.4            | ↓             |
| Deoxyhemoglobin               | ↓              | ↑             |
| Red cell Cl <sup>-</sup>      | ↓              | ↑             |
| Dissolved CO2                 | ↓              | ↑             |
| Carbamino hemoglobin          | ↓              | ↑             |

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## High Altitude

- Lower atmospheric pressure
- Lower pO<sub>2</sub>
- Hypoxia → hyperventilation
- ↓pCO<sub>2</sub> → respiratory alkalosis (pH rises)
- After 24-48hrs, kidneys will excrete HCO<sub>3</sub><sup>-</sup>
- pH will fall back toward normal

pH ↑  
pO<sub>2</sub> ↓  
pCO<sub>2</sub> ↓  
HCO<sub>3</sub><sup>-</sup> ↓

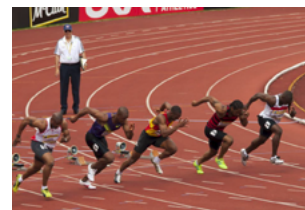


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## Exercise

- ↑ O<sub>2</sub> consumption
- ↑ CO<sub>2</sub> production
- ↑ Ventilation

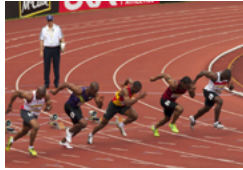


William Warby/Flickr

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## Exercise

- More  $\text{CO}_2$  produced by muscles
- $\text{CO}_2$  levels in *venous* blood rise
- More  $\text{O}_2$  consumed by muscles
- $\text{O}_2$  levels in *venous* blood fall

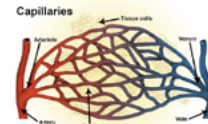


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## Exercise

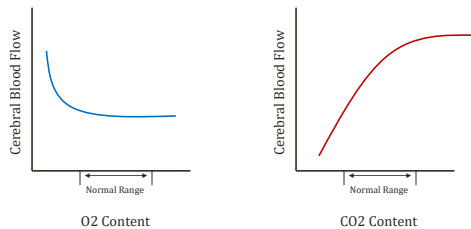
- $\uparrow$  ventilation and blood flow
- Normal  $\text{PaO}_2$  and  $\text{PaCO}_2$  despite metabolic changes

**Veins:  $\text{O}_2$  falls,  $\text{CO}_2$  rises**  
**Arteries:  $\text{O}_2$  and  $\text{CO}_2$  normal**



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## Cerebral Blood Flow



## Panic Attacks

- Hyperventilation
- Low  $\text{CO}_2$
- Hypocapnia  $\rightarrow$  cerebral vasoconstriction
- CNS symptoms (dizziness, blurred vision)



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## $\text{CO}_2$ and Breathing Control

- $\text{PaCO}_2$  is the major stimulus for breathing
- Central chemoreceptors in **medulla** most important
- Peripheral chemoreceptors: carotid and aortic bodies
  - Sense  $\text{CO}_2$  but more sensitive to  $\text{O}_2$
- High  $\text{PaCO}_2 \rightarrow$  increased respiratory rate
- Low  $\text{PaCO}_2 \rightarrow$  decreased respiratory rate



Medulla

## $\text{CO}_2$ and Breathing Control

- COPD patients: chronic retention of  $\text{CO}_2$ 
  - Lose sensitivity to  $\text{CO}_2$
  - Oxygen becomes major breathing stimulus
- Excess oxygen therapy given  $\rightarrow$  hypoventilation
- Theory: response to  $\text{CO}_2$  blunted
  - Respiratory depression with high  $\text{O}_2$
- New data indicates more complex
  - Haldane effect



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## CO<sub>2</sub> and Breathing Control

- CO<sub>2</sub> level useful to determine **ventilation status**
  - High CO<sub>2</sub>: hypoventilation
  - Low CO<sub>2</sub>: hyperventilation
- Clinical scenario:
  - Patient with pneumonia
  - O<sub>2</sub> applied via nasal cannula
  - O<sub>2</sub> level 95%
  - Blood gas: PaCO<sub>2</sub> = 60mmHg (high)

## CO<sub>2</sub> and Breathing Control

- Clinical scenario
  - Patient with neuromuscular disease (ALS)
  - O<sub>2</sub> saturation on O<sub>2</sub> 95%
  - Blood gas: PaCO<sub>2</sub> = 60 (high)
  - Respiratory muscles failing
- Symptoms of hypercapnia (high CO<sub>2</sub>)
  - Lethargy
  - Confusion
  - Agitation

# Lung Physical Exam

## Lung Physical Exam

Jason Ryan, MD, MPH



## Lung Exam

- Percussion
  - Finger against thorax → tap
- Auscultation
  - Stethoscope thorax
  - Upper, mid, lower lung fields
- Special techniques
  - Fremitus
  - Pectoriloquy



## Percussion

- Normal sounds = resonant
- Abnormal: dull or hyperresonant
- Dull
  - Pleural effusion
  - Consolidation (pneumonia)
- Hyperresonant → air trapped
  - Pneumothorax
  - Emphysema



## Lung Auscultation

- Normal breath sounds are vesicular
- Most all pathologic lung processes result in decreased lung sounds over affected area



## Adventitious Lung Sounds

- Rales
- Wheezes
- Rhonchi
- Bronchial breath sounds
- Stridor



## Rales

- Also called crackles
- Small airways “pop” open after collapse
- Early inspiratory, late inspiratory or expiratory
- Classic causes
  - Pulmonary edema (bases)
  - Pneumonia
  - Interstitial fibrosis



## Wheezes

- Air flows through narrowed bronchi
- Usually expiratory or inspiratory/expiratory
- Classic cause is asthma
- Other causes:
  - Heart failure (cardiac asthma)
  - Chronic bronchitis
  - Obstruction (tumor; localized wheeze)



## Rhonchi

- Secretions in large airways
- Coarse breath sounds
- Classic cause is COPD



## Bronchial Breath Sounds

- High pitched lung sounds
- Like flow through tube
- Longer expiratory phase than normal
- Seen in pneumonia with consolidation



## Stridor

- Wheeze that is almost entirely inspiratory
- Usually loudest over neck
- Indicates partial obstruction of larynx or trachea
- Some classic causes
  - Laryngotracheitis (croup)
  - Epiglottitis (Hib in children)
  - Retropharyngeal abscess
  - Diphtheria

## Pectoriloquy

- Sounds over chest through stethoscope
- Bronchophony
  - Voice sounds are louder and clearer
- Whispered pectoriloquy
  - Whispered "99-99-99"
  - Should be muffled
  - Abnormal if clear
- Egophony: "Eeeeeee" sounds like "Aaaay"
- All indicated fluid in lungs: Effusion, consolidation

## Fremitus

- Place hands on patients back
- Patient says "ninety-nine"
- Vibrations travel through airways to back
- Varies with density of lung tissue
- Only common condition with increased fremitus is lobar pneumonia
- Decreased in most other processes
  - Pleural effusion
  - Pneumothorax
  - Atelectasis



## Nail Clubbing

- Associated with many pulmonary diseases
- Bronchiectasis
- Cystic Fibrosis
- Lung tumors
- Pulmonary fibrosis
- Also cyanotic congenital heart disease



Image courtesy of James Heilman, MD



Image courtesy of Jfdwollf

# Pulmonary Function Tests

## Pulmonary Function Tests

Jason Ryan, MD, MPH

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## Dyspnea

- Many, many causes
- Deconditioning
- Cardiac causes
- Anemia
- Pulmonary causes

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## Pulmonary Dyspnea

- Obstruction
  - Can't get air out of lungs
  - Air trapped
  - Poor oxygenation
- Restriction
  - Can't get air into lungs
  - Poor oxygenation

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## Pulmonary Function Testing

- Determining flows, volumes in lung
- Helps determine cause of dyspnea
  - Sometimes unclear from history, exam, x-ray, etc.
- Helps determine disease severity/progression
  - Many diseases monitored by PFTs
  - COPD, Pulmonary Fibrosis

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## Spirometry

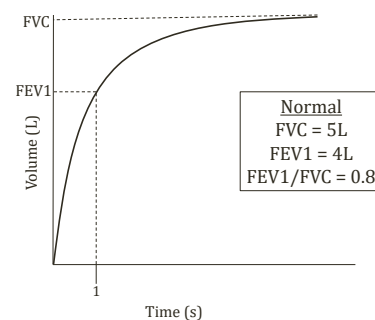
- Method for assessing pulmonary function
  - Pulmonary function tests (PFTs)
- Patient blows into machine
- Volume of air measured over time



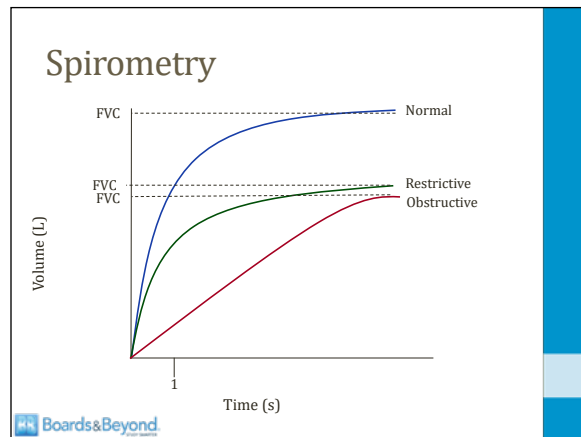
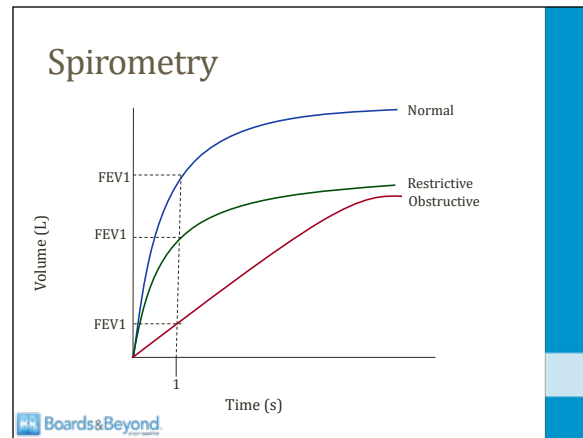
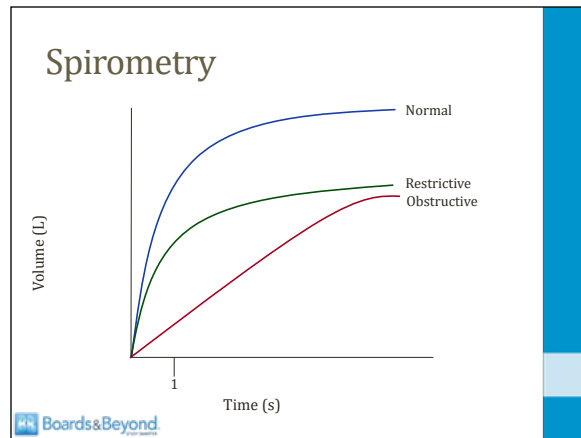
Image courtesy of Imatcha

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## Spirometry



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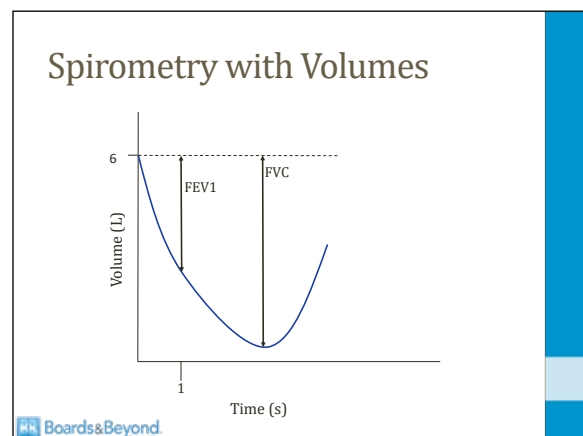
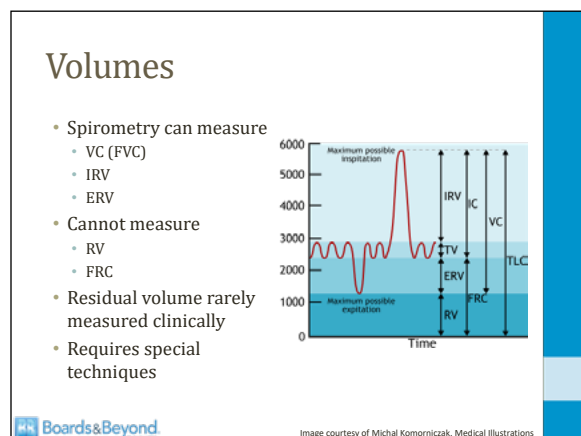


### Summary

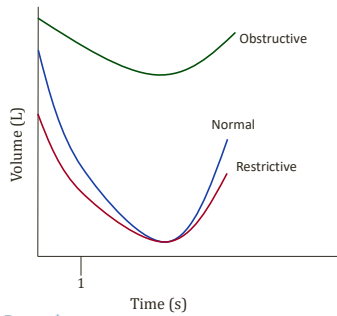
- FEV1 and FVC fall in both obstructive and restrictive diseases
- FEV1 falls MORE than FVC in obstructive

|             | FEV1 | FVC | FEV1/FVC |
|-------------|------|-----|----------|
| Obstructive | ↓↓   | ↓   | ↓        |
| Restrictive | ↓    | ↓   | >80%     |

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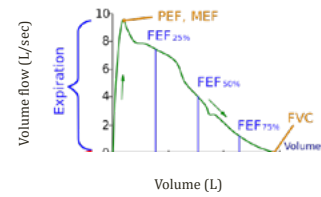


## Spirometry with Volumes



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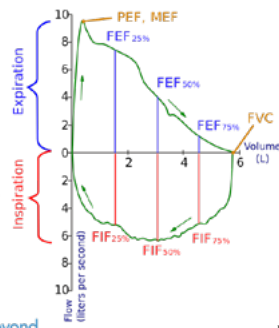
## Flow Volume Loop



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Image courtesy of SPhotographer

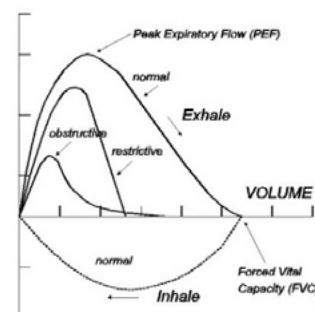
## Flow Volume Loop



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Image courtesy of SPhotographer

## Flow Volume Loops



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Image courtesy of Yasser Ammar

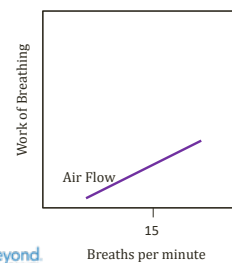
## Work of Breathing

- Work proportional to resistance

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## Work of Breathing

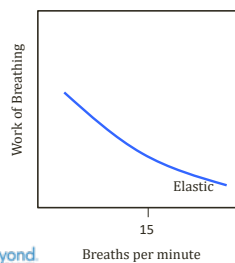
- Airflow resistance: Slower you breathe, less resistance



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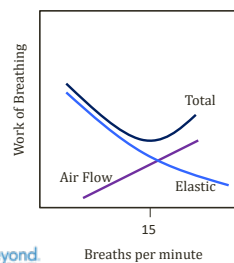
## Work of Breathing

- Elastic resistance: Faster you breathe, less resistance



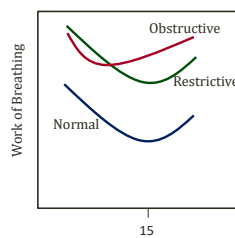
## Work of Breathing

- Slower you breathe, less airflow resistance
- Faster you breathe, less elastic resistance



## Work of Breathing

- Increases in obstructive and restrictive disease
- Different patterns



# Obstructive Lung Disease

## Obstructive Lung Disease

Jason Ryan, MD, MPH

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## Obstructive Lung Diseases

- Key points: Air trapping, slow flow out, less air out
- Reduced FEV1 (slow flow out)
- Reduced FVC (less air out)
- Reduced FEV1/FVC (hallmark)

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## Residual & Total Lung Volume

- Both go up in obstructive disease
  - From air trapping
- Both fall in restrictive disease
  - Less air fills the lungs due to restriction

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## Obstructive Lung Diseases

- Chronic bronchitis
- Emphysema
- Asthma
- Bronchiectasis

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## Chronic Bronchitis

- Chronic cough
- Productive of sputum
- At least 3 months over two years
- No other cause of cough present
- Strongly associated with smoking

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## Chronic Bronchitis

- Hypertrophy of mucous secreting glands
- Reid Index
  - Thickness of glands/total wall
  - >50% in chronic bronchitis
- Lungs can plug with mucous “mucous plugging”
- Increased risk of infection

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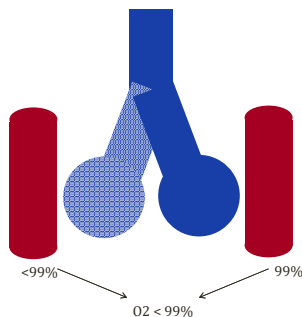
## Chronic Bronchitis

- Poor ventilation of lungs
- Increased CO<sub>2</sub>
- Decreased O<sub>2</sub>
- Hypoxic vasoconstriction
- Pulmonary hypertension
- Right heart failure (cor pulmonale)

## Chronic Bronchitis

- Cough
- Wheezing
- Crackles
- Dyspnea
- Cyanosis (shunting)

## Shunting



## Emphysema

- Smokers
  - Too many proteases created
  - Overwhelm anti-proteases
  - Upper lung damage
- $\alpha$ 1 anti-trypsin deficiency
  - Ineffective anti-proteases
  - Lower lobe damage



## Emphysema

- Destruction of alveoli
  - Smoke activates macrophages
  - Recruitment of neutrophils
  - Release of proteases
- Loss of elastic recoil
- Small airways collapse on exhalation
- Air "trapped" in lungs

## Emphysema

- Dyspnea
- Cough (less sputum than chronic bronchitis)
- Hyperventilation
- Weight loss
- Cor pulmonale
- Barrel Chest

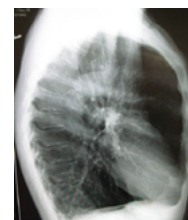
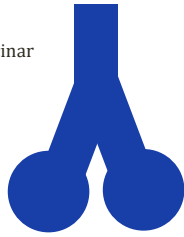


Image courtesy of James Helman, MD

## Acinus

- Acinus = bronchiole + alveoli
- Smokers = centriacinar damage
- $\alpha 1$  anti-trypsin deficiency = panacinar



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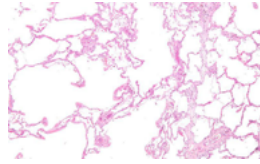


Image courtesy of Nephron

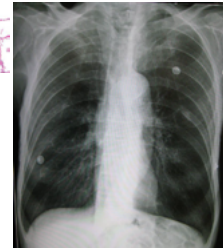
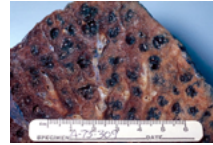
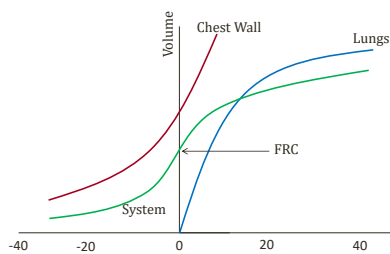


Image courtesy of James Hellman, MD

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## Chest Volumes and Pressures



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## Blue Bloater – Pink Puffer

- Chronic Bronchitis – Blue Bloater
  - Cyanosis from **shunting** (blue)
  - Air trapping (bloated)
- Emphysema – Pink Puffer
  - Loss of alveoli
  - Loss of surface area for O<sub>2</sub> absorption (**dead space**)
  - Hyperventilation to compensate (puffer)
  - Initially this maintains O<sub>2</sub> level (pink)

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## COPD

- Chronic Obstructive Pulmonary Disease
- Includes chronic bronchitis, emphysema, asthma
- Many similar symptoms (cough, dyspnea, wheezing)
- Many similar treatments

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## $\alpha 1$ Anti-trypsin Deficiency

- Inherited (autosomal co-dominant)
- Decreased or dysfunctional AAT
- AAT balances naturally occurring proteases
- Elastase found in neutrophils & alveolar macrophages

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## α1 Anti-trypsin Deficiency

- Lung
  - Panacinar emphysema
  - Imbalance between neutrophil elastase (destroys elastin) and elastase inhibitor AAT (protects elastin)
  - Lower lung damage
- Liver cirrhosis
  - Abnormal α1 builds up in liver
  - Only occurs in phenotypes with pathologic polymerization of AAT in endoplasmic reticulum of hepatocytes
  - Some patients have severe AAT deficiency but no intra-hepatocytic accumulation

## α1 Anti-trypsin Deficiency

- Classic case
  - Typical COPD symptoms: cough, sputum, wheeze
  - Younger patient (40s)
  - Imaging: emphysematous changes most prominent at bases
  - Obstructive PFTs
- Question often asks about panacinar involvement
- These patients should NEVER smoke
  - Stimulates neutrophil elastase production

## Asthma

- Reversible bronchoconstriction
- Usually due to allergic stimulus
  - Type I hypersensitivity reaction
- Airways are HYPERresponsive
- Common in children
- Associated with other allergic (atopic) conditions
  - Rhinitis, eczema
  - May have family history of allergic reactions

## Asthma Triggers

- URI
- Allergens (animal dander, dust mites, mold, pollens)
- Stress
- Exercise
- Cold
- Aspirin

## AERD

Aspirin Exacerbated Respiratory Disease

- Asthma, chronic rhinosinusitis, nasal polyposis
  - Chronic asthma/rhinosinusitis symptoms
  - Acute exacerbations after ingestion aspirin or NSAIDs
- Dysregulation of arachidonic acid metabolism
- Overproduction leukotrienes
- Treatment: Leukotriene receptor antagonists
  - Montelukast, Zafirlukast

## Asthma Symptoms

- Episodic symptoms
- Dyspnea, wheezing, cough
- Hypoxia during episodes
- Decreased I/E ratio
- Reduced peak flow
- Mucous plugging (airway obstruction/shunt)
- Death: Status asthmaticus



## Asthma Diagnosis

- Usually classic history/physical exam
- Methacholine challenge
  - Muscarinic agonist
  - Causes bronchoconstriction
  - Administer increasing amounts of nebulized drug
  - Spirometry after each dose
  - Look for dose at which FEV1 falls significantly
  - If dose is low → positive test

## Asthma Pathology

- Recurrent episodes
- Smooth muscle hypertrophy
- Inflammation

## Asthma Pathology

- Classic sputum findings
  - Curschmann's spirals
  - Charcot-Leyden crystals

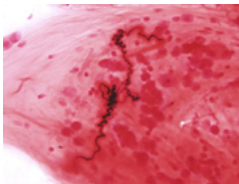


Image courtesy of Jnh649

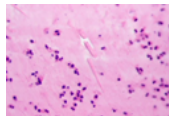


Image courtesy of Patho

## Pulsus Paradoxus

- Most frequent non-cardiac causes are asthma/COPD

## Bronchiectasis

- Result of chronic, recurrent airway inflammation
- Airways become permanently dilated
- Obstruction
  - Large airways dilated
  - Small/medium airways thickened bronchial walls

## Bronchiectasis



Image courtesy of Yale Rosen

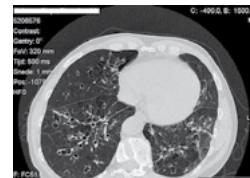


Image courtesy of Laura Fregonese, Jan Stolk

## Bronchiectasis Symptoms

- Recurrent infections
- Cough, excessive sputum (foul smelling)
- Hemoptysis
- Cor pulmonale
- Amyloidosis

## Bronchiectasis Etiologies

- Obstruction (tumor)
- Smoking
- Cystic fibrosis
- Kartagener's syndrome
- Allergic bronchopulmonary aspergillosis

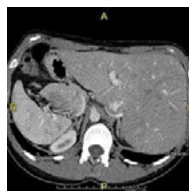
## Primary Ciliary Dyskinesia

Immotile-cilia syndrome

- Cilia unable to beat, beat normally, or absent
- Inherited (autosomal recessive)
- Gene mutations dynein structure/formation
- Dynein = motor protein creates movement

## Kartagener's syndrome

- Chronic sinusitis
- Bronchiectasis (chronic cough, recurrent infections)
- Male infertility
- Situs inversus



## Kartagener's syndrome

- Classic case:
  - Child
  - Recurrent sinus/ear infections
  - Chronic cough
  - Bronchiectasis on chest CT
  - Obstruction on PFTs
  - Situs inversus
- Question often asks about dynein protein

## ABPA

Allergic bronchopulmonary aspergillosis

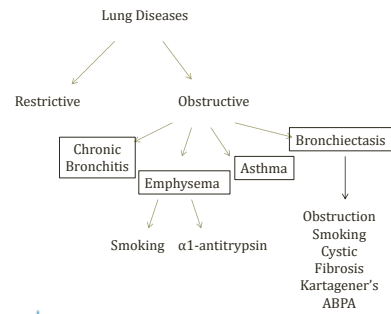
- Hypersensitivity (allergic) reaction to aspergillus
- Lungs become colonized with *Aspergillus fumigatus*
  - Low virulence fungus
  - Only infects immunocompromised or debilitated lungs
- Occurs predominantly in asthma and CF patients
- ABPA patients:
  - Increases Th2 CD4+ cells
  - Synthesis interleukins
  - Eosinophilia
  - IgE antibody production

## ABPA

Allergic bronchopulmonary aspergillosis

- Classic case
  - Asthma or CF patient
  - Recurrent episodes cough, fever, malaise
  - Brownish mucus plugs, hemoptysis
  - Peripheral blood eosinophilia
  - High IgE level
  - Bronchiectasis on imaging
  - PFTs with obstruction
- Diagnosis: Skin testing aspergillosis
- Treatment: Steroids

## Summary



# Restrictive Lung Disease

## Restrictive Lung Disease

Jason Ryan, MD, MPH



## Restrictive Lung Diseases

- Key points: Can't get air in → less air out
- Reduced FVC (less air in/out)
- Reduced FEV1 (less air in/out)
- Normal (>80%) FEV1/FVC (hallmark)



## Causes

- #1: Poor breathing mechanics
- #2: Interstitial lung diseases



## Poor Breathing Mechanics

- Not a primary pulmonary issue
- Under-ventilation of lungs
- Alveoli working: A-a gradient normal
- Neuromuscular
  - ALS, Polio, myasthenia gravis
- Structural
  - Scoliosis
  - Morbid obesity



## Interstitial Lung Disease



Bilateral, diffuse pattern  
Small, irregular opacities (reticulonodular)  
"Honeycomb" lung appearance.



Image courtesy of James Heilman, MD

## DLCO

Diffusing capacity in lung of carbon monoxide

- DLCO separates cases restrictive disease
- Restriction with normal DLCO
  - Extra-pulmonary cause: obesity
- Restriction with low DLCO
  - Interstitial lung disease



## DLCO

- DLCO = diffusing capacity of carbon monoxide
- Measures ability of lungs to transfer gas to RBCs
- Patient inhales small amount (not dangerous) CO
- CO uptake is diffusion limited
  - Amount taken up  $\approx$  diffusion function lungs
- Machine measures CO exhaled
- Normal = 75 – 140% predicted
- Severe disease <40% predicted

## Low DLCO Conditions

- Interstitial lung disease
- Emphysema
- Abnormal vasculature
  - Pulmonary hypertension
  - Pulmonary embolism
- Prior lung resection
- Anemia
  - Corrects when adjusted for Hb level

## Interstitial Diseases

- "Diffuse parenchymal lung diseases"
- Large group of disorders
- Similar clinical, radiographic, physiologic, or pathologic manifestations

## Interstitial Diseases

- Idiopathic pulmonary fibrosis
- Systemic diseases with interstitial lung features
  - Scleroderma
  - Rheumatoid arthritis
  - Goodpasture's
  - Wegener's
  - Sarcoidosis
- Pneumoconiosis
- Drug toxicity (amiodarone, methotrexate)
- Hypersensitivity pneumonitis

## Idiopathic pulmonary fibrosis

- Most common type: Idiopathic interstitial pneumonia
- Slow onset dyspnea
- Typically affects adults over the age of 40

## Pneumoconiosis

Occupational lung diseases

- Coal miner's lung
- Silicosis
- Asbestosis

## Coal miner's lung

- Inhalation of coal dust particles
- CXR or Chest CT:
  - Small, rounded, nodular opacities
  - Preference for the upper lobes

## Silicosis

- Inhalation of silica in quartz, granite, or sandstone
- Most widespread pneumoconiosis in US
- Foundries (metal production facilities)
- Sandblasting (abrasive blasting)
- Mines

## Silicosis

- Macrophages react to silica
- Inflammation → fibroblasts → collagen
- High prevalence of TB
  - Impaired macrophage killing
- High prevalence of bronchogenic carcinoma

## Silicosis

- Affects upper lobes
- Eggshell calcifications of lymph nodes

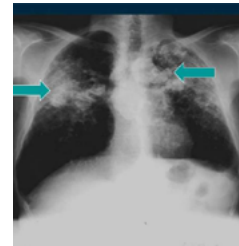


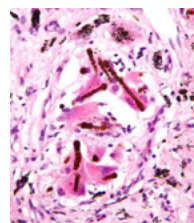
Image courtesy of Dr. Lucas Pedro Pablo Burchard Seifert

## Asbestosis

- Inhalation of asbestos fibers
- Shipbuilding, roofing, plumbing
- Classically affects lower lobes
- Three clinical problems:
  - Interstitial lung disease (asbestosis)
  - Pleural plaques
  - Lung cancer

## Asbestosis

- CXR: Calcified pleural plaques pathognomonic
- Path: Asbestos bodies (ferruginous body)
  - Asbestos fibers surrounded by a coating of iron and protein



## Asbestosis

- Bronchogenic carcinoma
- Mesothelioma
  - Asbestos is the only known risk factor for mesothelioma
  - Occurs decades after exposure
  - Pleural thickening and pleural effusion
  - Slow onset symptoms (dyspnea, cough, chest pain)
  - Poor prognosis

## Drug toxicity

- Bleomycin
- Busulfan
- Amiodarone
- Methotrexate

## Hypersensitivity pneumonitis

- Hypersensitivity reaction to environmental antigen
  - Agricultural dusts
  - Microorganisms (fungal, bacterial, or protozoa)
  - Chemicals
- Mixed type III/IV hypersensitivity
- Classic case is a farmer's lung
  - Moldy hay, grain exposure
- Also common in bird/poultry handlers
  - Waste from birds → dried, finely dispersed dust

## Hypersensitivity pneumonitis

- Classic case
  - Farmer or bird handler
  - Cough, dyspnea, chest tightness
  - Diffuse crackles
- Diagnosis (challenging):
  - Bronchoalveolar lavage
  - Inhalation challenge
  - Lung biopsy
- Treatment:
  - Avoid exposure
  - Steroids



# Treatment of COPD/Asthma

## Treatment of COPD & Asthma

Jason Ryan, MD, MPH



## COPD and Asthma Drugs

- Short-acting bronchodilators
  - Albuterol
  - Ipratropium
- Long-acting bronchodilators
  - Salmeterol, Formoterol
  - Tiotropium
- Steroids



## $\beta$ 2 Agonists

- Activate adenylate cyclase  $\rightarrow$   $\uparrow$ cAMP
- Relax bronchiole smooth muscle
- Short acting: Albuterol
  - Nebulizer or inhaler
  - Use during acute attacks (prn)
- Long acting: Salmeterol, Formoterol
  - Not used as monotherapy (always with ICS)
- Systemic side effects (rare)
  - Tremor, arrhythmia



## Muscarinic Antagonists

- Vagal nerve  $\rightarrow$  Ach  $\rightarrow$  Bronchoconstriction
- MA drugs block M receptors smooth muscle
- Prevents bronchoconstriction



## Muscarinic Antagonists

- Short acting: Ipratropium
- Long acting: Tiotropium



## Steroids

- Inhaled: Beclomethasone, Fluticasone, Budesonide
- Oral: Prednisone
- IV: Methylprednisolone (Solumedrol)



## Steroids

- Inhibit synthesis of cytokines
- Bind to glucocorticoid receptor (GR)
- Many, many immunosuppressive effects
- ↓ expression many interleukins, IFN- $\gamma$ , TNF- $\alpha$ , GM-CSF
- Inactivation NF-KB
  - Transcription factor
  - Induces production of TNF- $\alpha$

## Steroids

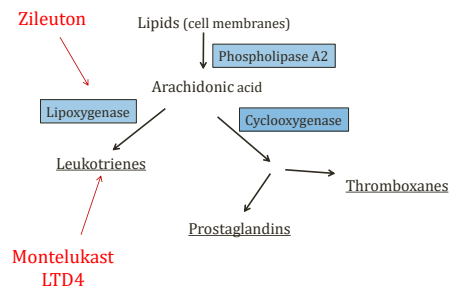
- Common side effect is oral candidiasis ("thrush")
- Patients instructed to rinse after inhalation



## Special Asthma Drugs

- Leukotriene receptor antagonists (PO)
  - Montelukast (Singulair)
  - Useful in aspirin sensitive asthma
- Zileuton (PO)
  - 5-lipoxygenase inhibitors
  - Blocks conversion of arachidonic acid to leukotrienes

## Eicosanoids



## Special Asthma Drugs

- Omalizumab (SQ injection)
  - IgG monoclonal antibody
  - Inhibits IgE binding to IgE receptor on mast cells & basophils
- Cromolyn (inhaler/nebulizer)
  - Inhibits mast cell degranulation
  - Blocks release of histamine, leukotrienes

## Theophylline

- Methylxanthines
- Multiple, complex mechanisms
- Bronchodilation
  - Likely through inhibition PDE
  - Less hydrolysis (breakdown) cAMP
  - ↑ cAMP
- Also down-regulates inflammatory cell functions

## Theophylline

- Narrow therapeutic index
- Levels must be monitored
- Dose must be titrated
- Goal is a peak serum concentration 10 to 20 mg/L

## Theophylline

- Metabolized by P450
- Many drug-drug interactions
- Common culprits:
  - Cimetidine
  - Ciprofloxacin
  - Erythromycin
  - Clarithromycin
  - Verapamil

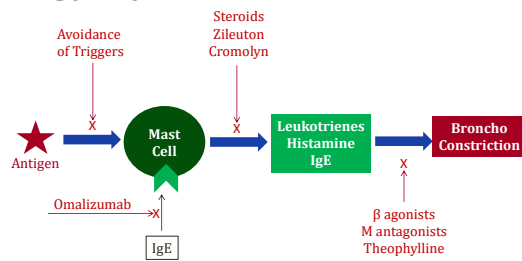
## Theophylline

- GI toxicity
  - Nausea, vomiting
- Neurotoxicity
  - Seizures
- Overdose scenario: Nausea, vomiting, seizures

## Theophylline

- Cardiotoxicity
  - Blocks adenosine receptors
  - Increased heart rate
  - Arrhythmias (atrial tachycardia, atrial flutter)
  - Cause of death in overdose/poisoning
- Key clinical scenario
  - Patient on theophylline for asthma/COPD
  - SVT
  - Adenosine fails to slow heart rate

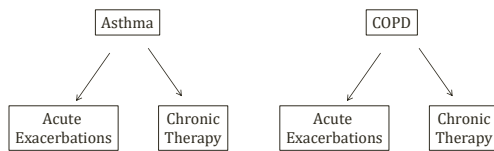
## Asthma



## Special COPD Drugs

- Theophylline
- Roflumilast (PO)
  - Phosphodiesterase-4 (PDE-4) inhibitor
  - Decreases inflammation
  - May relax airway smooth muscle

## Treatment Asthma & COPD



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## COPD: Acute Exacerbations

- Oxygen
- Nebulized albuterol +/- ipratropium (Combivent)
- IV or oral corticosteroids
  - Prednisone 60mg daily
  - Methylprednisolone 80mg IV q8hrs
- Antibiotics (severe, hospitalized patients)
  - Fluoroquinolones
  - Amoxicillin/clavulanate

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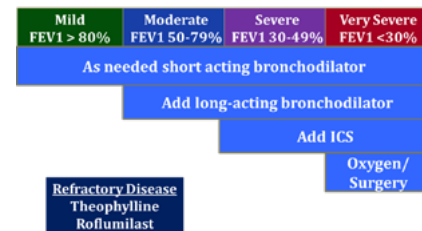
## GOLD Criteria

Global Initiative for Chronic Obstructive Lung Disease

| Stage  | Symptoms    | FEV1        |
|--------|-------------|-------------|
| Gold 1 | Mild        | FEV1 >80%   |
| Gold 2 | Moderate    | FEV1 50-79% |
| Gold 3 | Severe      | FEV1 30-49% |
| Gold 4 | Very Severe | FEV1 <30%   |

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## COPD: Chronic Therapy



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## COPD: Chronic Therapy

- Oxygen
  - Associated with increased survival
  - PaO2 < 55mmHG or O2 sat <88%
- Pulmonary rehabilitation
  - Improves exercise capacity, quality of life
  - Decrease dyspnea
- Vaccinations
- Smoking cessation

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## Asthma: Acute Exacerbations

- Oxygen
- Nebulized albuterol
- IV or oral corticosteroids
  - Prednisone 60mg daily
  - Methylprednisolone 80mg IV q8hrs
- Rarely used:
  - Ipratropium
  - IV Magnesium sulfate

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## Asthma: Chronic Therapy



## Surgical Treatment

- For advanced "end-staged" COPD
- Lung volume reduction surgery/Bullectomy
  - Remove diseased lung tissue
  - Allow healthy lung tissue more room to expand
- Lung transplant

# Pneumonia

## Pneumonia

Jason Ryan, MD, MPH

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## Pneumonia

- Infection of the lungs
- Three patterns
  - Lobar
  - Bronchopneumonia
  - Interstitial (atypical)

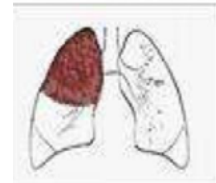
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## Lobar Pneumonia

- Classic form of pneumonia (*S. pneumoniae*)
- Bacteria acquired in nasopharynx
- Aerosolized to alveolus
- Enter alveolar type II cells
- Pneumococci multiply in alveolus
- Invade alveolar epithelium
- Pass from one alveolus to next (pores of Cohn)
- Inflammation/consolidation of lobes
- Can involve entire lung

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## Lobar Pneumonia



Images courtesy of Vijit Agrawal

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## Four Lobar Stages

- #1: Congestion (1<sup>st</sup> 24 hours)
  - Alveolar capillaries dilate
  - Exudate of bacteria develops
- #2: Red hepatization (2-3days)
  - Exudate of RBCs, neutrophils, fibrin
  - "Fresh" exudate: RBCs/WBCs intact
  - Pneumococci alive
  - Lobes look red

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## Four Lobar Stages

- #3: Gray hepatization (4-6days)
  - Gray, firm lobe
  - Exudate with neutrophils/fibrin
  - RBCs disintegrate
  - Dying pneumococci
- #4: Resolution
  - Return to normal (little scarring)
  - Enzymes digest exudate
  - Type II pneumocyte key for regeneration

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## Bronchopneumonia

- *Patchy* inflammation of multiple lobules
- Primary involvement airways and surrounding interstitium
- *Staphylococcus aureus*



Image courtesy of drahmed142010  
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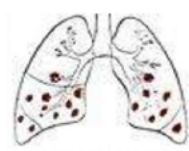


Image courtesy of Vijit Agrawal

## Interstitial Pneumonia

- Inflammatory infiltrate of alveolar walls only
- More indolent course
- Viruses
- *Legionella pneumophila*
- *Mycoplasma pneumoniae*
- *Chlamydia pneumoniae*

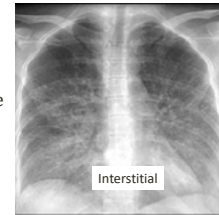


Image courtesy of drahmed142010

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## Atypical Pneumonia

- Pneumonia caused by:
  - *Legionella pneumophila*
  - *Mycoplasma pneumoniae*
  - *Chlamydia pneumoniae*
- Usually milder than strep pneumonia
- Respiratory distress rare
- Interstitial infiltrates on CXR
- "Walking pneumonia"



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Image courtesy of drahmed142010

## Causes of Pneumonia

Children

| Neonates<br><4weeks      | Children<br>4wk-18yr                                                            |
|--------------------------|---------------------------------------------------------------------------------|
| Group B Strep<br>E. Coli | Viruses (RSV)<br>Mycoplasma<br>Chlamydia Pneumoniae<br>Streptococcus Pneumoniae |

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## Causes of Pneumonia

Adults

- *S. pneumoniae* – most common
- *Haemophilus influenzae*
- *Mycoplasma pneumoniae*
- *C. pneumoniae*
- *Legionella*

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## Causes of Pneumonia

Adults

- Gram-negative rods
  - *Klebsiella*, *E. Coli*, *Pseudomonas*
  - Uncommon unless severe PNA
  - Often isolated in hospitalized patients
- *S. Aureus* (postinfluenza pneumonia)
- Anaerobes (aspiration PNA; lung abscess)
- Viruses
  - Influenza
  - RSV (children)

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## Signs/Symptoms

- High Fever
- Cough
- Sputum production
- Elevated WBC
- Pleuritic chest pain

## Diagnosis

- Usually:
  - History
  - Physical exam
  - X-ray (sometimes CT scan)
- Rarely
  - Sputum culture
  - Bronchoalveolar lavage

## Clinical Classes of Pneumonia

- Community acquired
  - Usually *S. Pneumoniae*, *H. Influenza*, *S. Aureus*
  - Sometimes *Mycoplasma*, *Chlamydia*, *Legionella* (atypicals)
- Nosocomial
  - Bad bugs
  - Often gram negatives (*Pseudomonas*, *Klebsiella*, *E. Coli*)
  - Hospital Acquired
  - Ventilator-associated pneumonia (VAP)
  - Healthcare-associated pneumonia (HCAP; nursing homes)

## Community Acquired PNA

Uncomplicated

- No co-morbidities
- No recent antibiotic use
- Low community rates resistance
- Azithromycin, Clarithromycin, or Doxycycline
- Three to five day course
  - Patient should be afebrile 48-72 hrs and clinically stable

## Community Acquired PNA

Complicated

- COPD, CKD, Diabetes, CHF, Alcoholism
- Recent antibiotic use
- Fluoroquinolone (levofloxacin)
- Amoxicillin plus azithromycin

## Nosocomial PNA

- Lots of resistance to antibiotics
- Gram negative rods
  - *E. coli*, *Klebsiella*, *Enterobacter*, *Pseudomonas*, *Acinetobacter*
- Staph Aureus including MRSA
- Often cover for pseudomonas, MRSA
- Sometimes multi-drug combinations
- Cefepime or Ceftazidime
- Imipenem or Meropenem
- Piperacillin-tazobactam (Zosyn)



## Complications

- Sepsis
- Respiratory failure
- Lung abscesses
- Pleural effusion
- ARDS

## ARDS

Acute Respiratory Distress Syndrome

- Triggered by various lung injuries
- Injury → release of pro-inflammatory cytokines
  - TNF, interleukins
- Cytokines recruit neutrophils to lungs
- Neutrophils release toxic mediators
  - Reactive oxygen species, proteases
- Damage to capillary endothelium and alveolar epithelium
- Protein escapes from vascular space
- Fluid pours into the interstitium

## ARDS

Triggers

- Sepsis (most common)
- Infection (PNA)
- Aspiration
- Trauma
- Acute pancreatitis
- Transfusion-related acute lung injury (TRALI)



Looks like pulmonary edema  
but PCPW is normal

## ARDS

Treatment

- Mechanical ventilation
- Low tidal volume
- Supportive care (fluids, nutrition)
- VAP pneumonia is serious complication

## Legionella

- First identified at American Legion convention
- Infection from inhalation of aerosolized bacteria
  - Not airborne
- Outbreaks at hotels with contaminated water
- Can cause nosocomial pneumonia in nursing homes

## Legionella

Symptoms

- Initially mild pneumonia symptoms
  - Fever; mild, slightly productive cough
- Can progress to severe pneumonia
- GI symptoms
  - Watery diarrhea, nausea, vomiting, and abdominal pain
- Hyponatremia ( $\text{Na} < 130 \text{ meq/L}$ ) common
  - Can occur in any PNA but more common Legionella

## Legionella

### Diagnosis

- Special culture requirements
  - Does not gram stain well
- Buffered charcoal yeast extract agar (BCYE)
- Iron and cysteine added for growth
- Supplemented with antibiotics and silver dyes
  - Antimicrobials prevent overgrowth by competing organisms
  - Dyes give distinctive color to Legionella
- Urinary antigen test
  - Rapid test available in minutes
  - Does not test for all Legionella types

## Legionella

### Diagnosis

- Classic Case
  - Mild cough
  - Watery diarrhea
  - Confusion (low Na)
  - Negative bacteria on gram stain
- Diagnose with urinary antigen test
- Treatment: Fluoroquinolone or Azithromycin

## Pontiac Fever

- Mild form of Legionella infection
- Fever, malaise, chills, fatigue, and headache
- No respiratory complaints
- Chest radiograph usually normal

## Mycoplasma Pneumonia

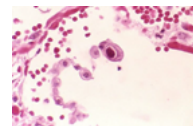
- Atypical pneumonia
- Can't see on gram stain (no cell wall)
- Classically causes outbreaks in young adults
  - College dorm residents
  - Military recruits
- CXR looks worse than symptoms
- Can cause autoimmune hemolytic anemia
  - IgM antibody → RBC antigen
  - "Cold" hemolytic anemia
- Stevens-Johnson syndrome

## Influenza Virus

- Atypical pneumonia
- Influenza A or B viruses
- Fever, headache, myalgia, and malaise
- Nonproductive cough, sore throat, runny nose
- Major complication is secondary pneumonia
  - Strep pneumoniae, Staph aureus, H. influenzae
  - Worsening symptoms after initial improvement
  - Cause of death

## CMV

- Pneumonia in transplant patients on immunosuppressive drugs
- "Owl eye" intranuclear inclusions



## RSV

### Respiratory Syncytial Virus

- Viral respiratory infection in infants
- Often seasonal outbreaks (Nov – April)
- Most common cause lower respiratory tract illness in children
  - Bronchiolitis, pneumonia, acute respiratory failure
- Often starts as upper airway infection
  - Runny nose
- Few days later, lower tract symptoms
  - Wheezing often present

## RSV

### Respiratory Syncytial Virus

- Treatment: Ribavirin
  - Inhibits synthesis of guanine nucleotides
- Prevention: Palivizumab
  - Monoclonal antibody against F protein
  - RSV contains surface F (fusion) protein
  - Causes respiratory epithelial cell fusion
  - Used in pre-term infants (high risk RSV)
  - Sometimes congenital heart disease

## RSV

### Respiratory Syncytial Virus

- Classic case
  - Young child (often <2yo)
  - Fever, runny nose
  - Few days later, cough, wheezing

## Aspiration Pneumonia

- Aspiration of microorganisms
- Bugs from oral cavity and nasopharynx to lungs
- Risk factors:
  - Reduced consciousness (anesthesia)
  - Seizures
  - Alcoholics
  - Dysphagia from neuromuscular weakness
- Classic patients:
  - Debilitated nursing home patient
  - Alcoholic

## Aspiration Pneumonia

- Klebsiella
- Staph Aureus
- Anaerobic bacteria
  - Peptostreptococcus
  - Fusobacterium
  - Prevotella
  - Bacteroides
- Clindamycin first-line therapy

## Klebsiella Pneumonia

- Can cause lobar pneumonia
- Often from aspiration
- Marked inflammation/necrosis
- Thick, mucoid and blood-tinged sputum
- "Currant jelly"

## Lung Abscess

- Contained, fluid-filled space in lungs
  - "Air fluid level" on imaging
- Usually a consequence of aspiration
- Rarely due to bronchial obstruction from cancer
- Predominantly anaerobes
  - Peptostreptococcus
  - Prevotella
  - Bacteroides
  - Fusobacterium
- Sometimes S. Aureus, Klebsiella
- Treatment: Clindamycin

## PCP

Pneumocystis jirovecii

- Diffuse interstitial pneumonia
- Requires immunocompromise
  - Classically HIV
  - AIDS-defining illness
- Yeast → inhaled
  - Usually no symptoms if immune system intact

## PCP

Pneumocystis jirovecii

- Diagnosed by microscopy
  - Sputum sample or BAL
- Staining required → cannot be cultured
- Special stains used
  - Silver stains often used

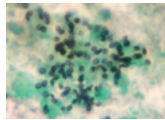


Image courtesy of Yale Rosen

## PCP

Pneumocystis jirovecii

- Treatments
  - TMP-SMX (first line)
  - Dapsone
  - Pentamidine
- Prophylaxis
  - TMP-SMX when CD4 <200cells/microL
  - High dose steroid or other immunosuppressant

# Pleural Disease

## Pleural Disease

Jason Ryan, MD, MPH



## What are the pleura?

- Two layers of tissue surrounding lungs
  - Visceral pleura – attached to lung
  - Parietal pleura – attached to chest wall
- Pleural space/cavity – between layers
- Pleural lined by mesothelial cells
- Secrete small amount pleural fluid for lubrication



## Pneumothorax

- Air in pleural space
- Two types to know about
  - Spontaneous
  - Tension



## Spontaneous PTX

- Primary
  - Rupture of subpleural bleb
  - Common in tall, thin young males
- Secondary
  - Older patients with underlying pulmonary disease
  - COPD

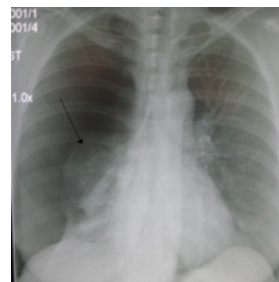


## Spontaneous PTX

- Sudden onset dyspnea
- Sometimes pleuritic chest pain
- CXR for diagnosis



## Pneumothorax



## Pneumothorax

### Treatment

- 100% Oxygen
  - Displaces nitrogen from capillary blood
  - ↑gradient for nitrogen reabsorption from pleural space
- Chest tube
  - Larger pneumothoraces (>15% lung volume)

## Tension PTX

- Usually from trauma
- Air enters pleural space but cannot leave
- Medical emergency
- Emergent thoracentesis/chest tube placement
- Trachea deviates AWAY from affected side

## Pleural Effusion

- Accumulation of fluid in pleural space



## Pleural Effusion

- Three general etiologies
  - Transudative
  - Exudative
  - Lymphatic

## Transudative Effusion

- Something driving fluid into pleural space
- Most commonly due to CHF (High pressure)
- Other causes:
  - Nephrotic syndrome (low protein)
  - Cirrhosis (low albumin)
- Mostly fluid in effusion
- Very little protein in effusion
- Usually treat for underlying cause (no drainage)

## Exudative Effusion

- Fluid leaking into pleural space
  - High vascular permeability
- Many causes
- Malignancy
- Pneumonia
- More protein in pleural fluid vs. transudative
- Usually requires drainage

## Transudate vs. Exudate

- Thoracentesis to obtain fluid sample
- Test for protein, LDH
- Light's Criteria – Exudate if:
  - Pleural protein/serum protein greater than 0.5
  - Pleural LDH/serum LDH greater than 0.6
  - Pleural LDH greater than 2/3 upper limits normal LDH

## Lymphatic Effusions

“Chylothorax”

- Lymphatic fluid effusion
- From thoracic duct obstruction/injury
- Malignancy most common cause
- Trauma (usually surgical)
- Milky-appearing fluid
- Very high triglycerides
  - TG usually > 110 mg/dL

## Other Effusions

- Hemothorax
  - High Hct in fluid
- Empyema
  - Infected pleural fluid
  - Pus, putrid odor, positive culture
- Malignant effusion
  - Positive cytology

## Mesothelioma

- Pleural tumor
- Asbestos is only known risk factor
  - Decades after exposure
- Imaging: Pleural thickening and pleural effusion
- Slow onset symptoms (dyspnea, cough, chest pain)
- Poor prognosis
  - Median survival 4 to 13 months untreated
  - 6 to 18 months treated with chemo

# Lung Cancer

## Lung Cancer

Jason Ryan, MD, MPH

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## Common Cancers

- Breast
- Prostate
- Lung (most deadly)
- Colorectal

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## Lung Cancer Risk Factors

- Cigarette smoking
  - Polycyclic Aromatic Hydrocarbons (PAHs)
- Radiation Therapy
  - Hodgkin's and breast cancer survivors
- Environmental toxins
  - Asbestos
  - Radon

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## Symptoms

- Usually advanced at presentation
- Cough, dyspnea, rarely hemoptysis
- Usually leads to chest imaging

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## Diagnosis

- Pulmonary nodule
- "Coin lesion"
- Compare with prior
- Biopsy for diagnosis

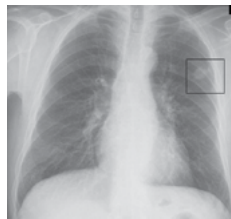


Image courtesy of Lange123

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## Benign Pulmonary Nodules

- Granulomas (80% benign nodules)
- Hamartomas
  - Lung tissue and cartilage (with scattered calcification)

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## Granulomas

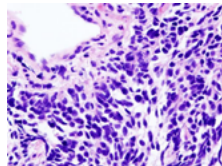
- Fungi
  - Histoplasmosis (patient from Midwest, Miss/Ohio river valley)
  - Coccidioidomycosis (southwest, California)
- Mycobacteria
  - Usually tuberculosis

## Lung Cancers

- Small cell (15%)
  - Fast growing; Early mets
  - Non amenable to surgical resection
  - Smokers
  - Treated with chemo
  - Poor prognosis
- Non-small cell (Most Common: 85%)
  - Can sometimes be resected
  - Better prognosis
  - Smokers and non-smokers

## Small Cell Cancer

- Poorly differentiated small cells
- Classic in male smokers
- Neuroendocrine tumor
- Central tumor



## Small Cell Cancer

### Paraneoplastic Syndromes

- ACTH
  - Cushing syndrome
  - Progressive obesity
  - Hyperglycemia
- ADH
  - SIADH
  - Hyponatremia (confusion)
- Antibodies
  - Antibodies against pre-synaptic Ca channels in neurons
  - Block release of acetylcholine
  - Lambert-Eaton syndrome
  - Main symptom is weakness

## Non-Small Cell Cancers

- Squamous Cell Carcinoma
- Adenocarcinoma
- Large cell carcinoma
- Bronchioloalveolar Carcinoma
- Carcinoid tumor

## Squamous Cell Carcinoma

- Hilar mass arising from bronchus
- Key pathology
  - Keratin production ("pearls") by tumor cells
  - Intercellular desmosomes ("intercellular bridges")
- Male smokers
- Can produce PTHrP
  - Hypercalcemia
  - Stones, bones, groans, psychiatric overtones
  - Bone and abdominal pain, confusion

## Adenocarcinoma

- Glandular tumor
- Most common lung cancer: nonsmokers/females
- Peripheral

## Large Cell Carcinoma

- Poorly differentiated
  - Lacks glandular or squamous differentiation
  - Lacks small cells
- Smokers cancer
- Central or peripheral
- Poor prognosis

## Bronchioloalveolar Carcinoma

- Subtype of adenocarcinoma
- Many similar features to adeno:
  - Nonsmokers, Peripheral
- Mucinous type: Derived from goblet cells
- Nonmucinous: Clara cells or type II pneumocytes
- Looks like PNA on CXR
  - Lobar consolidation
- Excellent prognosis
  - Surgery, radiotherapy, sometimes adjuvant chemotherapy

## Carcinoid tumor

- Neuroendocrine
- Well-differentiated cells
- Chromogranin positive
- Non-smokers
- Rarely causes carcinoid syndrome
  - Secretion of serotonin
  - Flushing, diarrhea

## Complications

- Pleural effusions
  - Tap fluid, send for cytology
- Phrenic nerve compression
  - Diaphragm paralysis
  - Dyspnea
  - Hemidiaphragm elevated on CXR
  - Sniff test
- Recurrent laryngeal nerve compression
  - Hoarseness

## SVC Syndrome

- Obstruction of blood flow through SVC
- Can be caused by compression from tumor
  - Lung Masses: NSCLC, SCLC
  - Mediastinal Masses: Lymphoma
- Other causes include thrombosis
  - Indwelling catheters, pacemaker wires
- Facial swelling or head fullness
- Arm swelling
- Can cause increased ICP
  - Headaches, confusion, coma
  - Cranial artery rupture

## SVC Syndrome

- Usually diagnosed CXR or CT Chest
- Various treatment options:
  - Anticoagulation for thrombus
  - Steroids (lymphoma)
  - Chemo/Radiation
  - Endovascular stenting

## Pancoast Tumor

- Carcinoma at apex of lung
- Involve superior sulcus
  - Groove formed by subclavian vessels
- Arm edema affected side
- Shoulder pain radiating toward axilla/scapula
- Arm paresthesias, weakness
- Can compress sympathetic nerves
- Horner's syndrome
  - Miosis
  - Ptosis
  - Anhidrosis



## Metastasis from Lung Cancer

- Adrenals
  - Usually found on imaging without symptoms
- Brain
  - Headache, neuro deficits, seizures
- Bone
  - Pathologic fractures
- Liver
  - Hepatomegaly, jaundice

## Metastasis to Lung

- More common than primary lung tumors
- Most commonly from breast or colon cancer
- Usually multiple lesions on imaging

# Sleep Apnea

## Sleep Apnea

Jason Ryan, MD, MPH

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## Sleep Apnea

- Apnea = cessation of breathing
- Sleep apnea = cessation of breathing during sleep
- Usually >10 seconds
- Multiple episodes per night are typical

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## Sleep Apnea Symptoms

- Unrestful sleep
- Daytime somnolence
- Loud snoring

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## Sleep Apnea Types

- Central sleep apnea
  - No effort to breathe
- Obstructive sleep apnea
  - Decreased air flow despite effort to breathe

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## Central Sleep Apnea

- Patients with marginal ventilation when awake
  - Hypoventilate when awake
  - Fall asleep → apnea periods
  - Central nervous system disease (encephalitis)
  - Neuromuscular diseases (polio, ALS)
  - Severe kyphoscoliosis
  - Narcotics

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## Central Sleep Apnea

- Cheyne-Stokes breathing
  - Cyclic breathing
  - Delayed detection/response to changes in  $\text{PaCO}_2$
  - Common in heart failure and stroke patients



Normal Breathing



Cheyne-Stokes

Images courtesy of Aleksa Lukic

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## Obstructive Sleep Apnea

- Recurrent soft tissue collapse in the pharynx
- Strongest risk factor is obesity

## Sleep Apnea Complications

- HTN
- Pulmonary HTN
- Arrhythmias
- Sudden death

## Erythropoiesis

- Chronic hypoxia
- EPO release

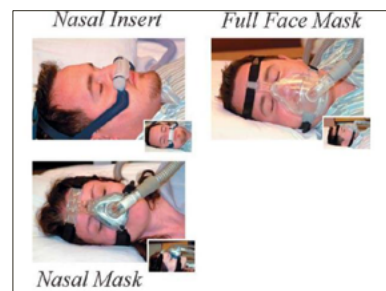
## Sleep Apnea Diagnosis

- Polysomnography
- "Sleep study"
- Patient sleeps in monitored setting
- EEG, eye movements
- O<sub>2</sub> level, HR, respiratory rate
- Number of apnea episodes recorded

## Sleep Apnea Treatments

- Weight loss
  - Takes time; not best option for exhausted patients
- CPAP
  - First line for symptomatic patients
- Upper airway surgery
  - Severe disease

## CPAP



# Cystic Fibrosis

## Cystic Fibrosis

Jason Ryan, MD, MPH

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## Cystic Fibrosis

- Inherited genetic disease
  - Autosomal recessive pattern
  - Both parents must be carriers
- Results in thick, sticky mucus in lungs/GI tract
- Common cause chronic lung disease in children

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## CFTR

Cystic Fibrosis Transmembrane Regulator

- CFTR protein is abnormal in CF
- CFTR gene encodes for the abnormal protein

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## CFTR

Cystic Fibrosis Transmembrane Regulator

- ATP ion transporter
- Epithelial Cell Functions
  - Pumps  $\text{Cl}^-$  out of epithelial cells
  - Against concentration gradient (uses ATP)
  - Creates a membrane potential that draws out  $\text{Na}/\text{H}_2\text{O}$
  - Hydrates mucosal surfaces (lungs, GI tract)
- Sweat gland functions
  - Removes  $\text{NaCl}$  from sweat (makes sweat hypotonic)
  - CF patients have high  $\text{NaCl}$  in sweat

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## CFTR Mutations

- Many mutations identified
- Most common mutation: delta F508
  - Deletion of 3 DNA bases
  - Codes for 508th AA acid: phenylalanine
- Most common consequence: abnormal processing
  - Abnormal protein folding
  - Prevents protein trafficking to correct cellular location

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## CF Pathophysiology

- Thick mucous in lungs
  - Recurrent pulmonary infections (*Pseudomonas*, *S. Aureus*)
  - Chronic bronchitis
  - Bronchiectasis
- Thick mucous in GI tract
  - Impaired flow of bile and pancreatic secretions
  - Malabsorption especially fats
  - Loss of fat soluble vitamins (A, D, E, K)
  - Steatorrhea

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## CF Presentation

- Usually diagnosed <2yo
- Respiratory disease (45%)
- Failure to thrive (28%)
- Meconium ileus (20%)

## CF Lung Disease

- Productive cough
- Hyperinflation of lungs on CXR
- Obstructive pattern
- Later disease
  - Chronic bronchitis
  - Bronchiectasis
- Acute exacerbations
- Pseudomonas aeruginosa: major pathogen in CF

## Pancreatic insufficiency

- Chronic pancreatitis
- CF-related diabetes
- Fat malabsorption
- Steatorrhea:
  - Frequent stools
  - Foul-smelling stools
  - Oily or greasy
  - Stools may float

## Pancreatic insufficiency

- Deficiencies of fat-soluble vitamins: A, D, E, and K
- Vitamin K: coagulopathy
- Vitamin D: rickets
- Vitamin A: Night blindness
- Vitamin E: Ataxia, hemolysis

## Meconium ileus

- Meconium
  - Meconium: first stool of newborn
  - Very thick and sticky
- Meconium ileus = bowel obstruction
  - Meconium too thick/sticky
  - Meconium plug forms
- Abdominal distension
- Vomiting
- Air fluid levels of X-ray
- Failure to pass meconium



Image courtesy of James Hellman, MD

## Other symptoms

- Biliary disease
  - Bile duct obstruction
  - Pale or clay colored stool
  - Elevation of LFTs
  - Hepatomegaly
  - Cirrhosis
  - Gallstones

## Other symptoms

- Infertility
  - 95 percent males with CF are infertile
  - Absent vas deferens
  - Problem is sperm transport not spermatogenesis
  - Can have children with assisted techniques

## Other symptoms

- Digital clubbing
- Nasal polyps



Image courtesy of James Heilman, MD

## Diagnosis

- Sweat chloride test
- Pilocarpine iontophoresis
- Pilocarpine gauze placed on skin
- Electrode placed over gauze
- Small electrical current drives pilocarpine into skin
- Sweating occurs
- Sweat collected on filter paper
- Chloride content analyzed
- High chloride level suggests CF
- DNA testing done if sweat test abnormal

## Diagnosis

- Rare CF patients have negative sweat test
  - Usually have milder disease
  - Often recurrent pulmonary and sinus infections
- If symptoms highly suggestive, can test nasal transepithelial potential difference
  - Measure nasal voltage
  - CF patients: more negative voltage
  - Due to abnormal sodium processing

## Treatment

- Promote clearance of airway secretions
  - Inhaled DNase (dornase alfa)
  - Inhaled saline
  - N-acetylcysteine
- Ivacaftor (tablets)
  - Increased chloride ion flux
  - Only for patients with G551D mutation
- Exacerbations are treated with antibiotics
- Lung transplantation

## Other Treatments

- Pancreatic enzyme replacement
- Vitamins (A, D, E, K)
- Vaccinations



## Prognosis

- Average life span ~ 37 years
- Death from lung complications

## Screening

- Prenatal
  - Test for 23 most common CF mutations in US
  - Often test mother first and stop if negative
- Newborn
  - ↑ blood levels immunoreactive trypsinogen (IRT)
  - Blood test → if positive → sweat test

# Tuberculosis

## Tuberculosis

Jason Ryan, MD, MPH

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## Tuberculosis

- Ancient disease: Found in mummies!
- Old name: Consumption
- Tubercle = round nodule
- Tuberculosis = multiple round nodules

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## Mycobacterium tuberculosis

- Obligate aerobes
  - Prefer lungs
  - Reactivation disease prefers upper lobes
- Facultative intracellular pathogens
  - Infect macrophages

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## Culture of TB

- Difficult to culture
  - Special media used
  - Lowenstein-Jensen agar
- Slow growing
- Does not stain well with Gram stain
- This is due to mycolic acids in cell wall
  - Also fatty acids and complex lipids

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## Acid Fast

- Cell walls impermeable to many dyes
- Stain with very concentrated dyes plus heat
  - Lipid soluble; contain phenols
- Once stained, plate rinsed with acid decolorizer
  - "Acid fast stain"
- TB resists decolorization with acid solvents
- Some other bacteria (Nocardia) also do this

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## Virulence Factors

- Trehalose dimycolate ("cord factor")
  - Helps evade immune response
  - Causes granuloma formation
  - Triggers cytokine release
- Sulfatides
  - Glycolipids
  - Inhibits fusion of phagosomes/lysosomes
- Catalase-peroxidase
  - Resists host cell oxidation

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## Spread of TB

- Spreads through the air
- Active TB patient's cough, sneeze, etc.
- Inhaled by uninfected person
- Can spread rapidly in crowded areas

## Exposure to TB

- Most patients will not develop active disease
  - Infection can clear or remain "latent"
- Small proportion patients develop active disease

## Primary TB

### Clinical Picture

- Mainly a disease of childhood or chemo patients
  - Ineffective immune response
- Gradual onset: weeks
- Fever
- Cough
- Pleuritic chest pain
- Fatigue, arthralgias

## Primary TB

### Pathophysiology

- First week
  - TB infects macrophages
  - Phagocytosed
  - Intracellular bacterial proliferation

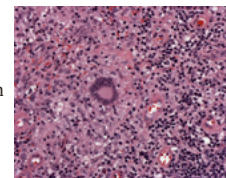
## Primary TB

### Pathophysiology

- Two to four weeks
  - Cell-mediated immune system controls TB
  - TH1 response
  - Activation of CD4+ T cells
  - Interferon- $\gamma$  secreted
  - Activated macrophages and cytotoxic T lymphocytes

## Granulomas

- Granulomatous inflammation
- Caseating necrosis
- Macrophages transform to :
  - Epithelioid cells
  - Langhans giant cells
- Fibroblasts activated  $\rightarrow$  collagen
- T-cell mediated delayed type hypersensitivity reaction
  - Type IV hypersensitivity reaction



## Hilar Lymphadenopathy

- CXR often normal
- Classic finding is hilar lymphadenopathy
- Occur as early as 1 week after infection
- Resolve slowly over months to years

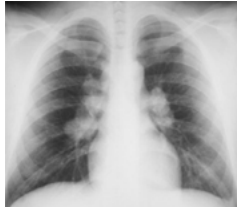
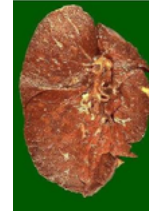
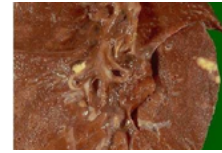


Image provided by www.learningradiology.com, courtesy of Dr. William Herring, MD, FACC. Used with permission.

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## Ghon Foci

- Ghon foci form
  - Granulomas
  - Subpleural
  - Mid to lower lungs
- Ghon foci plus lymph node is Ghon complex
- Calcified Ghon complex is a Ranke complex



Images courtesy of kazimer

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## Primary TB Resolution

- Most (90%) patients control infection
  - Disease heals leaving fibrosis
  - Sometimes completely clears
  - Usually enters latent phase ("walled off")
  - Immunity develops
  - PPD positive
- Rare (10%) patients have expanded illness
  - Miliary dissemination
  - More common with HIV, CKD, DM (impaired immunity)

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## Miliary TB

- Hematogenous spread of TB
- Progressive primary infection or reactivation
- Nearly any organ system can be involved
  - Bones
  - Liver
  - CNS (meningitis)
  - Heart (pericarditis)
  - Skin



Image courtesy of Yale Rosen

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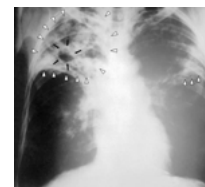
## Miliary TB

- Pott's disease
  - Spine infection (osteomyelitis)
  - Back pain, fever, night sweats, weight loss
- Constrictive pericarditis

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## Reactivation TB

- Reactivation of dormant TB
- Cough, weight loss, fatigue
- Fever
- Night sweats
- Chest pain
- Often cavitation (caseous and liquefactive necrosis)
- Hemoptysis (erode pulmonary vasculature)
- CXR classically shows upper lobe lesions



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## Reactivation TB

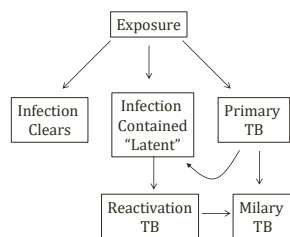
- Can occur when immune compromise develops
- HIV infection
- TNF- $\alpha$  inhibitors
  - Used in autoimmune diseases
  - Etanercept, Infliximab
- Diabetes

## Aspergilloma

Aspergillus  
ABPA  
Invasive Aspergillosis  
Aspergilloma

- Fungus ball
- Caused by Aspergillus fumigatus
- Non-invasive form of aspergillosis
- Grows in pre-formed cavities
- Pulmonary TB is most common association
- Often asymptomatic
- Can cause hemoptysis
- Diagnosis: Imaging plus sputum culture
- Treatment: Observation vs. surgery

## TB Infection Summary



## Diagnosis of Active TB

- Usual method: 3 sputum samples
  - Usually about 8hrs apart
  - Spontaneous or induced
  - Induced: Inhalation of aerosolized saline by nebulizer
- Acid-fast smear and culture

## Diagnosis of Active TB

- Not necessary to hospitalize just for TB suspicion
- Outpatients: Remain at home, avoid visitors, mask
- Inpatients: Respiratory isolation
  - Private room
  - Negative air pressure
  - Persons entering must wear a respirator
  - Tight seal over the nose and mouth

## Diagnosis of Latent TB

- Identification of latent TB crucial to infection control
- Diagnosis: Tuberculin skin testing (TST)
  - SC injection purified protein derivative (PPD)
    - 5 tuberculin units (0.1 mL)
- Wait 48 hours
- Measure diameter of induration (not erythema)

## PPD Testing

| Induration | Interpretation                                  |
|------------|-------------------------------------------------|
| <5mm       | Negative                                        |
| >5mm       | Positive if HIV, Immunosuppressed               |
| >10mm      | High risk individuals*                          |
| >15mm      | Healthy patients >4yo with low likelihood of TB |

\* Silicosis, CKD, DM, IV drug users, homeless, prison employees, others

## PPD Testing

- False negatives can occur
- Immunosuppressive drugs
  - Corticosteroids
  - TNF- $\alpha$  inhibitors
- Immunocompromised
  - HIV
  - CKD
  - Malnutrition
- Diseased lymph system
  - Sarcoidosis
  - Some lymphomas or leukemias

## BCG Vaccine

- Bacille Calmette-Guérin
- Live strain of *Mycobacterium bovis*
- More effective in patients with no TB exposure
  - About 80% effective in children
  - Less effective in adults
- Used in children in areas with high prevalence of TB
- Creates false positive PPD

## Treatment of Positive PPD

- Most patients with latent TB will not develop disease
- Small proportion may reactivate
- Prophylaxis lowers risk
- Commonly isoniazid (INH) for 9 months
- Further PPD testing not indicated
  - Will remain positive for life

## Treatment of Active TB

- Requires multi-drug regimens
- Typical regimen:
  - Isoniazid
  - Rifampin
  - Pyrazinamide
  - Ethambutol
  - Sometimes streptomycin
- Sometimes direct observation therapy (DOT)
- Risk of Multi-drug resistant (MDR) TB

## Isoniazid

- Blocks synthesis of mycolic acids
- Bacteria lose their acid fastness
- katG-encoded catalase-peroxidase
  - Converts INH to active form
  - Mutations lead to INH resistance
  - Monotherapy produces resistance

## Isoniazid

- Neurotoxic
  - Neuropathy, ataxia, and paresthesia
  - Competes with B6 as co-factor neurotransmitter synthesis
  - Pyridoxine (B6) co-administered
  - Limits neurotoxicity
- Hepatotoxic (check LFTs)
  - Probably related to metabolites of INH
- Drug-induced lupus

## Rifampin

- Inhibit bacterial DNA-dependent RNA polymerase
- Blocks RNA synthesis
- Main side effects are liver, GI
  - Increased LFTs
  - GI upset: nausea, cramps, diarrhea
- Red/orange discoloration fluids (not dangerous)
  - Urine
  - Saliva
  - Sweat, tears
  - CSF

## Rifampin

Other uses

- Leprosy
- Meningococcal prophylaxis
- Chemoprophylaxis in contacts of children HiB

## Pyrazinamide

- Mechanism unknown
  - Converted to pyrazinoic acid (PZA)
  - May be more active in acidic environment inside macrophages
- Hepatotoxic
  - Can raise LFTs
- Competes with uric acid for excretion in kidneys
  - Can raise uric acid levels
  - Hyperuricemia
  - Gout exacerbations

## Ethambutol

- Inhibits arabinosyl transferase
  - Polymerizes arabinose for mycobacteria cell walls
- Key side effect: optic neuropathy
  - Red-green color blindness
  - Difficulty discriminating red and green hues
  - Loss of visual acuity
  - Reversible

## Streptomycin

- Older, aminoglycoside drug
- Inhibits bacterial 30S ribosomal subunit
  - Prevents protein synthesis
- Lots of resistance
  - Mutations of genes for ribosomal proteins

## Tuberculosis Key Points

- Mycolic acid cell walls → acid fast
- Infects macrophages (intracellular)
- Delayed type hypersensitivity reaction
- Hilar lymphadenopathy; Ghon complex
- Reactivation in upper lobes (immunosuppressed)
- Latent infection diagnosed with PPD
- Treat latent disease with INH
- Treat active disease with multidrug regimen



# Sarcoidosis

## Sarcoidosis

Jason Ryan, MD, MPH



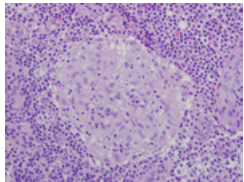
## Sarcoidosis

- Granulomatous disease
  - Granulomas form many places in the body
- Immune-mediated
  - Immune cells play major role
- Unknown cause



## Sarcoidosis

- Hallmark is widespread non-caseating granulomas
- Tightly packed central area of macrophages, epithelioid cells, multinucleated giant cells
- Surrounded by lymphocytes, monocytes, mast cells, fibroblasts



## Pathology

- Cell mediated immune process
- Accumulation of TH1 CD4+ helper T cells
  - High CD4:CD8 ratio
- Secrete IL-2 and interferon- $\gamma$
- IL-2 stimulates TH1 proliferation
- IFN- $\gamma$  activates macrophages
- Ultimately leads to granuloma formation
- Key players: CD4 T cells, IL-2, IFN-  $\gamma$



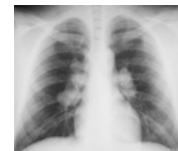
## Organ Involvement

- Lungs (most common)
- Skin
- Eye
- Heart
  - Conduction disease (heart block)
  - Cardiomyopathy
- Many other systems rarely involved
  - Renal: Renal failure
  - CNS: Neurosarcoid, Bells Palsy, Motor loss
- Any system can be involved



## Lung Involvement

- Classic finding is hilar lymphadenopathy
- Classic symptom is cough, dyspnea
- Can cause infiltrates
- Can cause pulmonary fibrosis



## Skin Involvement

- Many lesions possible
  - Plaques, maculopapules, subcutaneous nodules
- Classic lesion is erythema nodosum
  - Inflammation of fat cells under skin
  - Tender red nodules
  - Usually on both shins

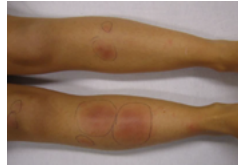
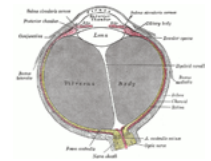


Image courtesy of James Heilman, MD

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## Eye Involvement: Uveitis

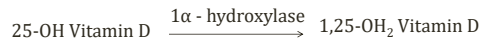
- Can involve many parts of eye
- Classic is uveitis
- Uvea:
  - Iris, ciliary body, choroid
- Uveitis Types
  - Anterior (iris, ciliary)
  - Posterior (choroid)
- Often mild symptoms
  - Dry eye, blurry vision
- Often detected on routine exam



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## Other Sarcoidosis Features

- Hypercalcemia
  - Elevated 1- $\alpha$  hydroxylase activity in alveolar macrophages
  - Increased vitamin D levels (calcitriol)
- High ACE levels
  - Non-specific finding
  - Elevated in many lung diseases



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## Classic Presentation

- African American female
- Hilar lymphadenopathy
- Cough, dyspnea
- Often asymptomatic, detected on routine chest x-ray

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## Treatment

- Steroids
- Other immunosuppressants
  - Methotrexate
  - Azathioprine
  - Mycophenolate

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# Pulmonary Embolism

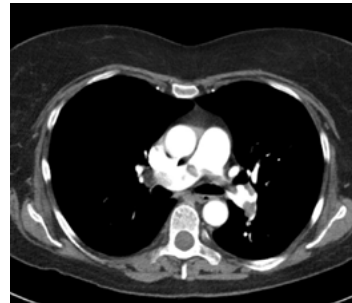
## Pulmonary Embolism

Jason Ryan, MD, MPH

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## Pulmonary Embolism

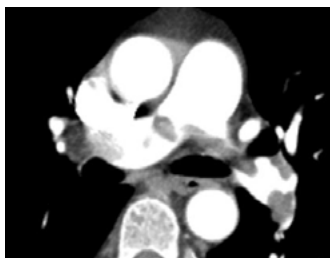
CT Angiogram



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## Pulmonary Embolism

CT Angiogram



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## Pulmonary Embolism

- Thrombus in pulmonary artery
- Rarely formed in heart or pulmonary vasculature
- Majority come from **femoral vein or deep leg veins**
- Travels to lung via IVC → RA → RV

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## Pulmonary Embolism

- Can be “unprovoked”
- Often secondary to a **hypercoagulable state**
  - Secondary: Malignancy, surgery, etc.
  - Primary: Protein C/S deficiency, ATIII deficiency, etc.

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## Pulmonary Embolism

- **Chest pain**
  - Classic presentation is pleuritic
- **Respiratory distress**
  - Dyspnea
  - Hypoxemia
  - Tachypnea
- Massive PE can cause **sudden death**
  - Obstruction to flow through pulmonary arteries
- Small, chronic emboli: **pulmonary hypertension**

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## Pulmonary Embolism

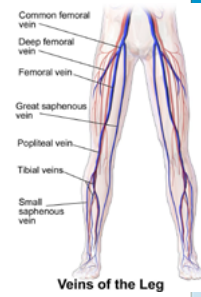
Ventilation-Perfusion

- Dead space
  - Ventilation without perfusion
- **V/Q mismatch**
- Hyperventilation
- Blood gas findings variable
- Classic findings: **low PaO<sub>2</sub> and low PCO<sub>2</sub>**

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## Deep Vein Thrombosis

- Thrombus within a deep vein
- Usually occurs in calf or thigh
- Commonly femoral/popliteal veins
- Can extend or "grow"
- Precedes pulmonary embolism
- Often 2° hypercoagulable state



Veins of the Leg

Bruce Blaus/Wikipedia

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## Deep Vein Thrombosis

- Often asymptomatic until PE
- Calf pain
- Palpable cord (thrombosed vein)
- Unilateral edema
- Warmth, tenderness, erythema
- Homan's sign: calf pain with dorsiflexion of foot
- Diagnosis: **Lower extremity ultrasound**



James Hellman, MD

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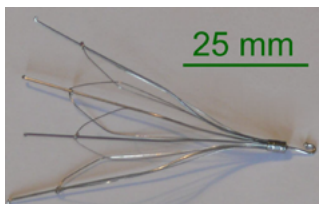
## Deep Vein Thrombosis

- Similar treatment to PE
  - "DVT/PE"
  - "Venous thromboembolism" (VTE)
- Prevention important in **hospitalized patients**
  - Hypercoagulable
  - Immobility, stasis of blood, inflammation
- **Prophylaxis:** SQ heparin, LMWH

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## IVC Filter

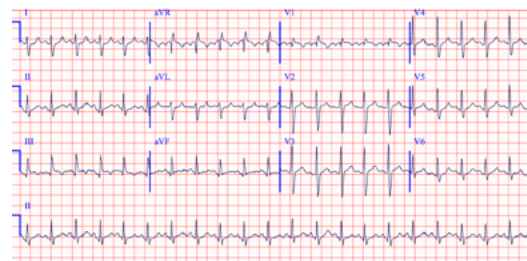
- Used in high-risk DVT patients
- Placed to prevent pulmonary embolism



BozMo/Wikipedia

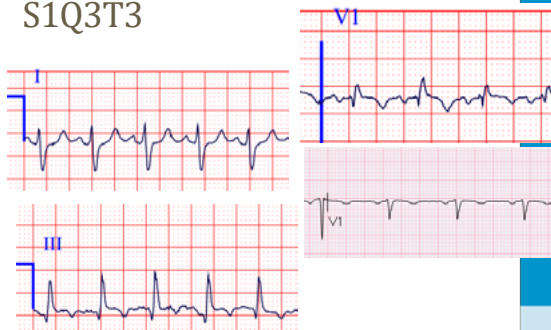
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## S1Q3T3



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## S1Q3T3



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## Wells Score

|                                         |    |
|-----------------------------------------|----|
| Active cancer                           | 1  |
| Immobilization of the lower extremities | 1  |
| Recently bedridden                      | 1  |
| Localized tenderness                    | 1  |
| Leg swelling                            | 1  |
| One leg swollen > other                 | 1  |
| Pitting edema                           | 1  |
| Superficial veins visible               | 1  |
| Alternative diagnosis likely            | -2 |

Score  $\geq 3$  High Probability  
1-2 Mod Probability  
0 Low Probability

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## D-dimer

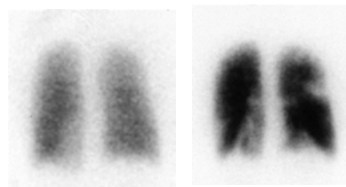
- Degradation product of fibrin
- Sensitive but not specific (unidirectional)
  - Levels elevated in DVT/PE
  - Levels also elevated in many, many other conditions
- Useful when normal in setting of low-mod Wells score

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## Pulmonary Embolism

### Diagnosis

- CT angiogram
- VQ Scan



Westgate EJ, FitzGerald GA  
Pulmonary Embolism in a Woman Taking Oral Contraceptives and Valdecoxib.  
PLoS Medicine Vol. 2, No. 7, e197. doi:10.1371/journal.pmed.0020197

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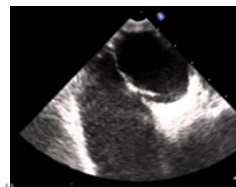
## Treatment DVT/PE

- Initial treatment with heparin or LMWH
- Transition to warfarin (oral)
- Massive PE: thrombolysis (tPA)

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## Patent Foramen Ovale

- Found in ~25% adults
- Failure of foramen ovale to close after birth
- Can allow venous clot to reach arterial system (brain)
- Rarely causes **stroke** in patients with DVT/PE



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## Fat Embolism

- Often occurs after a **long bone fracture**
- Fat may cross lungs → small artery infarctions
- Fat embolism syndrome: pulmonary, neuro, skin



Hellerhoff / Wikipedia

## Fat Embolism

- **Lung**
  - Dyspnea, hypoxemia
  - Diffuse capillary leak (ARDS)
  - Often requires mechanical ventilation
- **Neurological**
  - Usually **confusion**
  - May develop focal deficits
- **Petechiae**



James Heilman, MD/Wikipedia

## Amniotic Fluid Embolism

- During labor or shortly after
- Amniotic fluid, fetal cells, fetal debris enter maternal circulation
- Inflammatory reaction
- Often fatal



Wikipedia/Public Domain

## Amniotic Fluid Embolism

- **Phase I**
  - Pulmonary artery vasospasm → pulmonary hypertension
  - Right heart failure
  - Hypoxia
  - Myocardial capillary damage → left heart failure
  - Pulmonary capillary damage → ARDS
  - Acute respiratory distress syndrome
- Key features: **respiratory distress, ↓O<sub>2</sub>, hypotension**

## Amniotic Fluid Embolism

- **Phase II (hemorrhagic phase)**
  - Massive hemorrhage
  - DIC
- Key feature: **bleeding**
- **Seizures** also often occur

# Chest X-rays

## Chest X-rays

Jason Ryan, MD, MPH

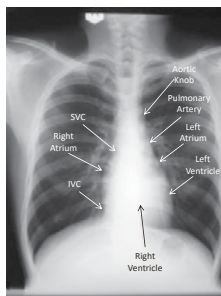


## Chest X-ray

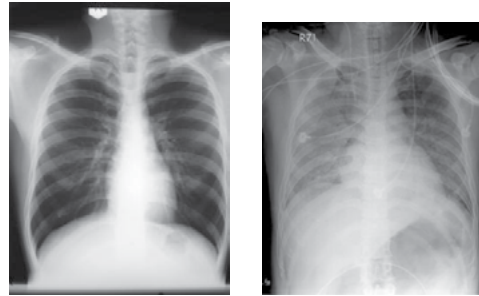
- Difficult to see different structures
- Many, many normal variants
- Many, many pathologic findings
- Reasonable goals:
  - Basic chest anatomy
  - Classic examples of pathology



## Chest Anatomy



## Pulmonary Edema



## Pulmonary Edema



Image courtesy of Radiology Assistant

## Pleural Effusion

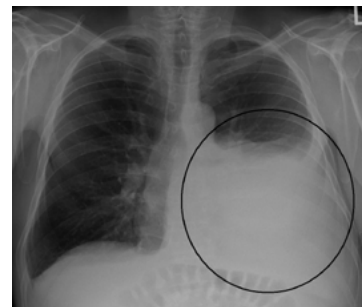
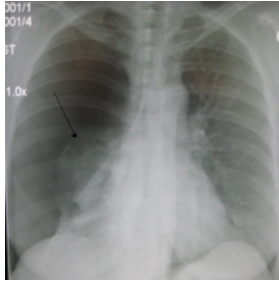


Image courtesy of James Heilman, MD

## Pneumothorax



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## Lobar Pneumonia



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## Interstitial Fibrosis



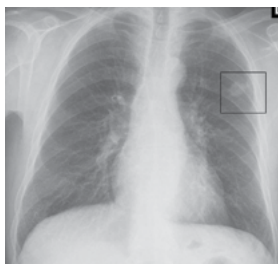
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## Hilar Lymphadenopathy



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## Pulmonary Nodule



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Image courtesy of Lange123